

# A $Q$ -OPERATOR MODEL FOR CYCLOTOMIC SYNTOMIC TATE TWISTS

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ABSTRACT. Let  $p$  be an odd prime and  $R = \mathbf{Z}_p[\zeta_p]$ . We identify  $\mathbf{Z}_p(i)(R)$ , for  $i \geq 1$ , with the total fiber of an explicit commuting square of  $q$ -operators on  $A = \mathbf{Z}_p[[q-1]]$ . The comparison applies Liu's one-correspondence construction to the defining Nygaard pullback and removes the Hodge-completed derived de Rham term by octahedral cancellation. A finite Herr reduction and an integral cyclotomic residue then compute the cohomology without using the syntomic-étale comparison.

## 1. INTRODUCTION

Fix an odd prime  $p$ . Throughout the paper,

$$\begin{aligned} A &= \mathbf{Z}_p[[q-1]], & d &= [p]_q = \frac{q^p - 1}{q - 1}, & R &= A/(d) = \mathbf{Z}_p[\zeta_p], \\ \varphi(q) &= q^p, & \gamma(q) &= q^{p+1}, & h &= q(q-1), & s &= p+1, \\ E(T) &:= [p]_{1+T}, & \pi &= \zeta_p - 1, & e &= E'(\pi) \in R. \end{aligned}$$

Thus  $E(T)$  means the substitution  $q = 1 + T$  in  $[p]_q$ , not a product. For  $n \geq 1$ , write  $[n]_x = (x^n - 1)/(x - 1)$ . Put

$$\lambda(q) = \frac{s}{[s]_q}, \quad g = \frac{\gamma(d)}{d}, \quad r_i(a) = \lambda(q)^i \gamma(a), \quad \tilde{r}_i(a) = \lambda(q^p)^i \gamma(a).$$

The power series  $[s]_q$  is a unit in  $A$ . Define the additive operators

$$(1.1) \quad D_i = \frac{r_i - \text{id}}{h}, \quad \tilde{D}_i = \frac{\tilde{r}_i - \text{id}}{dh}, \quad F_i = \varphi - d^i, \quad V_i = q^{p-1}\varphi - d^{i-1}.$$

We use the convention  $\mathbf{Z}_p(i)(R) = \text{fib}(\widehat{\varphi}\{i\} - \text{can})$ , where the map runs from  $\mathbb{N}^{\geq i} \widehat{\mathbb{A}}_R^{(1)}\{i\}$  to  $\widehat{\mathbb{A}}_R^{(1)}\{i\}$ .

**Theorem 1.1.** *For every  $i \geq 1$ , after fixing the cyclotomic coordinate  $q$ , the orientation  $d = [p]_q$ , and hence the induced oriented trivialization of the Breuil–Kisin Tate line, there is an equivalence of  $p$ -complete  $\mathbf{Z}_p$ -complexes canonical relative to precisely these choices,*

$$(1.2) \quad \mathbf{Z}_p(i)(R) \simeq \text{TotFib} \begin{array}{ccc} A & \xrightarrow{D_i} & A \\ F_i \downarrow & & \downarrow V_i \\ A & \xrightarrow{\tilde{D}_i} & A. \end{array}$$

Equivalently, the right side is the three-term complex

$$(1.3) \quad C_i^\bullet = \left[ A \xrightarrow{a \mapsto (D_i(a), F_i(a))} A \oplus A \xrightarrow{(b, c) \mapsto V_i(b) - \tilde{D}_i(c)} A \right]$$

in degrees 0, 1, 2. Its cohomology, as  $\mathbf{Z}_p$ -modules, is

$$\begin{aligned} H^0(C_i^\bullet) &= 0, \\ H^1(C_i^\bullet) &\cong R \oplus \mathbf{Z}_p/(s^i - 1), \\ H^2(C_i^\bullet) &\cong \begin{cases} 0, & i = 1, \\ \mathbf{Z}_p/(s^{i-1} - 1), & i \geq 2. \end{cases} \end{aligned}$$

The isomorphism in degree one is a noncanonical isomorphism of  $\mathbf{Z}_p$ -modules.

The proof begins with the pullback description of the Nygaard filtration from [1, Definition 6.6]. The adjacent Nygaard cofiber sequence of [2, Corollary 5.2.8] turns this pullback into an octahedral fiber in which the Hodge-completed derived de Rham summand cancels. Liu's one-correspondence then realizes the remaining prismatic coefficient as a two-term arithmetic complex. Passing to the central fiber of the Rees construction reduces the associated-graded calculation to Pham's Weyl model. The resulting maps are equivalences in every filtration degree, and completeness reconstructs the filtered complex.

Under this comparison, the canonical map is the inclusion of Nygaard lattices, while divided Frobenius gives the operators  $F_i$  and  $V_i$  in the two cochain degrees. Their difference gives the vertical arrows in (1.2). Section 4 then computes the cohomology of the total complex by a finite Herr reduction and an integral residue calculation.

The comparison proved here concerns the Breuil–Kisin Tate line. AKN provides the Nygaard and syntomic package, whereas the arithmetic two-term realization comes from Liu's cyclotomic one-correspondence. The argument neither requires a classification of arbitrary coefficients on the Nygaard stack nor appeals to syntomic–étale comparison.

## 2. THE FILTERED NYGAARD COMPARISON

**2.1. Arithmetic differential lines.** Liu writes  $\epsilon$  for the generator of the infinitesimal direction in the cyclotomic one-correspondence. We denote it by  $\epsilon_{\text{ar}}$  in order to distinguish the arithmetic line  $A\epsilon_{\text{ar}}$  from the other square-zero directions used below. Gros–Le Stum–Quirós define two free rank-one modules of differentials

$$\Omega_{A/\mathbf{Z}_p}^\Delta = A d_\Delta q, \quad \Omega_{A/\mathbf{Z}_p}^{q^p} = A d_{q^p} q$$

in [4, §2]. The scalar derivation on the first line is

$$\partial_\Delta = \frac{\gamma - \text{id}}{h},$$

which agrees with Liu's scalar operator [3, Construction 3.1.10 and Proposition 3.1.21]. Hence there is a unique  $A$ -linear isomorphism compatible with scalar derivations,

$$(2.1) \quad \iota_\Delta : A\epsilon_{\text{ar}} \xrightarrow{\sim} \Omega_{A/\mathbf{Z}_p}^\Delta, \quad \epsilon_{\text{ar}} \longmapsto d_\Delta q.$$

The Bockstein comparison is the isomorphism

$$(2.2) \quad \iota_{q^p} : dA\epsilon_{\text{ar}} \xrightarrow{\sim} \Omega_{A/\mathbf{Z}_p}^{q^p}, \quad d\epsilon_{\text{ar}} \longmapsto d_{q^p} q,$$

and in the chosen bases it is expressed by

$$(2.3) \quad d_{q^p} q = d d_\Delta q.$$

Thus the square

$$\begin{array}{ccc} A\epsilon_{\text{ar}} & \xrightarrow[\sim]{\iota_{\Delta}} & \Omega_{A/\mathbf{Z}_p}^{\Delta} \\ \times d \downarrow & & \downarrow \text{Bock} \\ dA\epsilon_{\text{ar}} & \xrightarrow[\sim]{\iota_{q^p}} & \Omega_{A/\mathbf{Z}_p}^{q^p} \end{array}$$

commutes. We shall write

$$\omega = d\epsilon_{\text{ar}} = d_{q^p}q, \quad v = e\epsilon_{\text{ar}}.$$

The first element belongs to the divided Nygaard differential line; the second is the basis of the arithmetic line obtained after finite descent.

*Remark 2.1.* The element  $v$  need not coincide with an integral logarithmic generator in the Hodge–Tate splitting of Bhatt–Lurie. The proof uses that splitting only as an augmented equivalence under its degree-zero summand. Lemma 2.7 shows that changing the basis of the shifted summand leaves the four maps in (1.2) unchanged.

**2.2. The Nygaard pullback and its logarithmic residue.** We write  $R^{\Delta} = (\text{Spf } R)^{\Delta}$  and  $R^{\text{HT}} = (\text{Spf } R)^{\text{HT}}$  throughout. For a fixed  $i \geq 1$ , write

$$\mathcal{H}_i = \mathcal{H}_{\Delta}^{(1)}(R/\mathbf{Z}_p)\{i\}.$$

For every integer  $r \geq 0$ , set

$$\begin{aligned} N_{r,i} &= \mathbf{N}^{\geq r} \text{R}\Gamma(\text{WCart}, \mathcal{H}_i), \\ P_{r,i} &= \text{R}\Gamma(\text{WCart}, \mathbf{N}^{\geq r} F^* \mathcal{H}_i), \\ D_r &= \text{Fil}_{\text{Hod}}^{\geq r} \widehat{\text{dR}}_{R/\mathbf{Z}_p}, \\ H_r &= D_r \widehat{\otimes}_{\mathbf{Z}_p} \text{R}\Gamma(\text{WCart}^{\text{HT}}, \mathcal{O}_{\text{WCart}^{\text{HT}}}). \end{aligned}$$

Definition 6.6 of [1] is the cartesian square

$$(2.4) \quad \begin{array}{ccc} N_{r,i} & \longrightarrow & D_r \\ \downarrow & & \downarrow \\ P_{r,i} & \longrightarrow & H_r. \end{array}$$

The lower-right corner is the full Hodge-completed derived de Rham complex; it is not replaced by  $R$ .

Bhatt–Lurie’s Hodge–Tate calculation [2, Proposition 3.5.11 and Example 3.5.12] gives, for every quasi-coherent coefficient  $\mathcal{E}$ , the canonical fiber sequence

$$\text{R}\Gamma(\text{WCart}^{\text{HT}}, \mathcal{E}) \longrightarrow \mathcal{E}_{\eta} \xrightarrow{\Theta_{\mathcal{E}}} \mathcal{E}_{\eta}.$$

For the structure sheaf, the Leibniz rule forces  $\Theta_{\mathcal{O}} = 0$ . In a stable category the fiber of the specified zero morphism  $\mathbf{Z}_p \rightarrow \mathbf{Z}_p$  is canonically  $\mathbf{Z}_p \oplus \mathbf{Z}_p[-1]$ , and its structural map to  $\mathbf{Z}_p$  is the first projection. Thus this is a natural splitting in the enhanced derived category, not a splitting chosen only on cohomology. Their Example 3.5.12 identifies the generator of the shifted summand with the logarithm. Consequently there is a canonical equivalence

$$(2.5) \quad \text{R}\Gamma(\text{WCart}^{\text{HT}}, \mathcal{O}_{\text{WCart}^{\text{HT}}}) \simeq \mathbf{Z}_p \oplus \mathbf{Z}_p[-1].$$

The unit is  $(\text{id}, 0)$ . Indeed, its composite with the first projection is the identity, while its possible shifted component lies in  $\text{Hom}_{\mathcal{D}(\mathbf{Z}_p)}(\mathbf{Z}_p, \mathbf{Z}_p[-1]) = 0$ . Thus it has no component in the fiber of that

projection. The second basis vector is the logarithm of the tautological  $\mathbf{G}_m^\sharp$ -character. Write the lower map in (2.4), under (2.5), as

$$\beta_{r,i} = (c_{0,r}, \text{Res}_r) : P_{r,i} \longrightarrow D_r \oplus D_r[-1].$$

**Lemma 2.2** (The logarithmic form of the AKN pullback). *There is a canonical equivalence*

$$N_{r,i} \simeq \text{fib}(P_{r,i} \xrightarrow{\text{Res}_r} D_r[-1]).$$

*Proof.* Under (2.5), the right vertical map in (2.4) is

$$(\text{id}, 0) : D_r \longrightarrow D_r \oplus D_r[-1].$$

Thus the defining pullback is the fiber of

$$D_r \oplus P_{r,i} \longrightarrow D_r \oplus D_r[-1], \quad (x, y) \longmapsto (x - c_{0,r}(y), -\text{Res}_r(y)).$$

The source automorphism

$$T : D_r \oplus P_{r,i} \xrightarrow{\sim} D_r \oplus P_{r,i}, \quad T(x, y) = (x + c_{0,r}(y), y),$$

conjugates this morphism to

$$\text{id}_{D_r} \oplus (-\text{Res}_r) : D_r \oplus P_{r,i} \longrightarrow D_r \oplus D_r[-1].$$

Fibers commute with finite direct sums in a stable category. Its fiber is therefore

$$\text{fib}(\text{id}_{D_r}) \oplus \text{fib}(-\text{Res}_r) \simeq 0 \oplus \text{fib}(\text{Res}_r),$$

which proves the asserted equivalence.  $\square$

### 2.3. Adjacent Nygaard cancellation.

2.3.1. *The abstract octahedral cancellation.* The following formal lemma is the operation which removes the full Hodge corner. It does not divide a map by  $d$ .

**Lemma 2.3** (Octahedral cancellation). *Let  $\mathcal{C}$  be a stable infinity-category and fix a composable pair*

$$X \xrightarrow{v} Y^- \xrightarrow{u} Y.$$

*Put  $P = \text{fib}(uv)$  and  $D = \text{cofib}(u)$ . Let*

$$\alpha_{u,v} : P \longrightarrow D[-1]$$

*be the third arrow in the canonical fiber sequence supplied by the octahedral axiom,*

$$\text{fib}(v) \longrightarrow \text{fib}(uv) \xrightarrow{\alpha_{u,v}} \text{cofib}(u)[-1].$$

*Then*

$$\text{fib}(\alpha_{u,v}) \simeq \text{fib}(v).$$

*Proof.* Apply the fiber form of the octahedral axiom to the displayed composable pair

$$X \xrightarrow{v} Y^- \xrightarrow{u} Y.$$

It gives a fiber sequence

$$\text{fib}(v) \longrightarrow \text{fib}(uv) \longrightarrow \text{cofib}(u)[-1].$$

By definition its second arrow is  $\alpha_{u,v}$ ; taking its fiber proves the claim. Notice that this construction does not use, and in general there is no, a projection  $\text{fib}(uv) \rightarrow Y[-1]$ .  $\square$

We now apply the lemma without replacing the pointwise Nygaard object by a relative two-term complex. Let

$$\mathcal{E}_{r,i} := N^{\geq r} F^* \mathcal{H}_i \in \mathcal{D}(\mathrm{WCart})$$

denote the pointwise Nygaard-filtered prismatic crystal occurring in Definition 6.6 of [1]. For  $r = 0$ , put  $\mathcal{E}_{-1,i} = \mathcal{E}_{0,i}$ ; the filtration is exhaustive in nonpositive degrees, but the adjacent arrow

$$\mathcal{I} \otimes \mathcal{E}_{-1,i} \longrightarrow \mathcal{E}_{0,i}$$

still contains multiplication by the Cartier–Witt ideal.

Write

$$\pi_R : R^\Delta \longrightarrow \mathrm{WCart}, \quad \rho : \mathrm{Spf}(A) \longrightarrow R^\Delta$$

for the forgetful map and Liu’s cyclotomic  $q$ -prism chart. Set

$$M_{r,i} := \rho^* \pi_R^* \mathcal{E}_{r,i}, \quad M_{r,i}^- := \rho^* \pi_R^* (\mathcal{I} \otimes \mathcal{E}_{r-1,i}).$$

The chosen orientation  $\rho^* \mathcal{I} = (d)$  gives a canonical identification  $M_{r,i}^- \simeq dM_{r-1,i}$ , where

$$dM_{r-1,i} := (d) \otimes_A M_{r-1,i}.$$

The structural arrow from this oriented tensor copy to  $M_{r,i}$  is induced by multiplication by  $d$ ; no injectivity assertion is implicit in the notation.

Here and below we use Liu’s cyclotomic parameters  $\alpha = 0$  and  $n = \infty$ . The standing assumption that  $p$  is odd gives  $p > 2$ , which is exactly the hypothesis needed in Liu’s Proposition 3.3.1 and Theorems 3.5.2–3.5.3. Liu’s Proposition 3.1.21 attaches to each  $M_{r,i}$ , functorially, the operator associated with the arithmetic correspondence. The coefficient is pulled back from  $\mathrm{WCart}$ , so this construction is equivariant for  $G = \mathbf{F}_p^\times$ . Let  $\mathcal{L}_{\mathrm{ar}} = A\epsilon_{\mathrm{ar}}$  be Liu’s arithmetic line, and set

$$\begin{aligned} X_{r,i} &= M_{r,i}^G, \\ Y_{r,i} &= (M_{r,i} \otimes_A \mathcal{L}_{\mathrm{ar}})^G, \\ X_{r,i}^- &= (M_{r,i}^-)^G, \\ Y_{r,i}^- &= (M_{r,i}^- \otimes_A \mathcal{L}_{\mathrm{ar}})^G. \end{aligned}$$

Liu’s finite full-faithfulness theorem [3, Theorem 3.5.2] first identifies the global sections of  $\mathcal{E}_{r,i}$  with those of its pullback to  $[R^\Delta/G]$ . His absolute one-correspondence theorem [3, Proposition 3.3.1] then computes the latter by the arithmetic operator, and therefore

$$(2.6) \quad P_{r,i} \simeq \mathrm{fib}(X_{r,i} \xrightarrow{\partial_{r,i}^{\mathrm{ar}}} Y_{r,i}).$$

Denote this natural equivalence by

$$\lambda_{r,i} : P_{r,i} \xrightarrow{\sim} \left[ X_{r,i} \xrightarrow{\partial_{r,i}^{\mathrm{ar}}} Y_{r,i} \right],$$

where the two-term complex denotes the fiber of the displayed arrow. Ordinary invariants may be used because  $|G| = p - 1$  is a unit in  $\mathbf{Z}_p$ . More explicitly, if  $\bar{\pi}_R : [R^\Delta/G] \rightarrow \mathrm{WCart}$  is induced by  $\pi_R$ , then full faithfulness gives, naturally in  $\mathcal{F}$ ,

$$\mathrm{R}\Gamma(\mathrm{WCart}, \mathcal{F}) = \mathrm{RHom}_{\mathrm{WCart}}(\mathcal{O}, \mathcal{F}) \xrightarrow{\sim} \mathrm{RHom}_{[R^\Delta/G]}(\mathcal{O}, \bar{\pi}_R^* \mathcal{F}) = \mathrm{R}\Gamma([R^\Delta/G], \bar{\pi}_R^* \mathcal{F}).$$

Thus this global-sections comparison uses only full faithfulness of pullback, not essential surjectivity. The scope of the cited results is important here. We record the applicability check explicitly. Liu’s Proposition 3.1.21 constructs the operator functor on the whole derived category  $\mathcal{D}(X_n^\Delta)$ . Proposition 3.1.23 constructs the canonical map from global sections to its fiber. Proposition 3.3.1 is stated for every object of that category, for  $n \in \mathbf{N} \cup \{\infty\}$ , and proves that this map is an equivalence; it imposes no perfectness, boundedness, or finite-generation hypothesis on the coefficient. At  $n = \infty$ , pullback to  $\mathrm{Spf} A$  is understood in Liu’s derived  $(p, d)$ -complete category, which is also the completion

convention for pullback of a quasi-coherent complex from  $R^\Delta$ . His Theorem 3.5.2 is fully faithful on the whole category  $\mathcal{D}(\mathrm{WCart}_n)$ , and Theorem 3.5.3 applies to every object of  $\mathcal{D}(\mathrm{WCart}^{\mathrm{HT}})$ . By construction,  $\mathcal{E}_{r,i}$  and  $\mathcal{I} \otimes \mathcal{E}_{r-1,i}$  are objects of  $\mathcal{D}(\mathrm{WCart})$ , while  $D_r \widehat{\otimes} \mathcal{O}_{\mathrm{WCart}^{\mathrm{HT}}}$  is an object of  $\mathcal{D}(\mathrm{WCart}^{\mathrm{HT}})$ . Thus the cited results apply to these three objects with exactly the stated completion convention. All adjacent arrows are morphisms in those same stable infinity-categories, so no closure of a smaller coefficient class under cofibers is being assumed below. Put

$$\Lambda_{\mathrm{ar}} := (R\epsilon_{\mathrm{ar}})^G = \mathbf{Z}_p v, \quad v = e\epsilon_{\mathrm{ar}}, \quad D_r^{\mathrm{ar}} := D_r \widehat{\otimes}_{\mathbf{Z}_p} \Lambda_{\mathrm{ar}}.$$

Also put

$$P_{r,i}^- := \mathrm{R}\Gamma(\mathrm{WCart}, \mathcal{I} \otimes \mathcal{E}_{r-1,i}).$$

The superscript ar distinguishes this integral arithmetic coordinate from the logarithmic coordinate in the Bhatt–Lurie splitting.

The adjacent relative Nygaard map of Bhatt–Lurie [2, Corollary 5.2.8, Remark 5.5.2, and Construction 5.5.3] gives a functorial cofiber sequence of quasi-coherent complexes on  $\mathrm{WCart}$ . After tensoring with Liu’s arithmetic line and taking finite invariants, it gives cofiber sequences

$$(2.7) \quad X_{r,i}^- \longrightarrow X_{r,i} \longrightarrow D_r, \quad Y_{r,i}^- \xrightarrow{u_{r,i}} Y_{r,i} \longrightarrow D_r^{\mathrm{ar}}.$$

Here the definition of  $Y_{r,i}^-$  includes multiplication by the prism ideal. At  $r = 0$ , the preceding Nygaard filtration is exhaustive, but this multiplication by  $d$  remains part of the adjacent map; therefore  $Y_{0,i}^-$  is the  $d$ -multiple of the unfiltered arithmetic line.

### 2.3.2. Liu realization of the adjacent cofiber.

**Lemma 2.4** (Exact naturality and specified nullhomotopies). *Let  $F, G : \mathcal{C} \rightarrow \mathcal{D}$  be exact functors between stable infinity-categories, and let  $\eta : F \Rightarrow G$  be a natural equivalence. If a cofiber sequence in  $\mathcal{C}$  is regarded as a  $\Delta^2$ -diagram, including its specified nullhomotopy of the composite, then  $\eta$  gives an equivalence between the two resulting  $\Delta^2$ -diagrams in  $\mathcal{D}$ . It preserves that nullhomotopy. The same remains true after applying functorial augmentations and taking their cofibers.*

*Proof.* A cofiber sequence is an object of the full subcategory of  $\mathrm{Fun}(\Delta^2, \mathcal{C})$  cut out by the cofiber condition; its nullhomotopy is the 2-simplex of that object, not an additional equality in the homotopy category. Exactness of  $F$  and  $G$  preserves the cofiber condition, while naturality of  $\eta$  gives an objectwise equivalence of the resulting  $\Delta^2$ -diagrams and hence preserves their 2-simplices. Finally, cofiber is an exact functor on the arrow category, so the assertion persists after the stated augmentations.  $\square$

**Lemma 2.5** (The Hodge quotient has zero arithmetic operator). *Under the canonical Hodge–Tate trivialization in AKN Definition 6.6, the pullback of  $D_r \widehat{\otimes} \mathcal{O}_{\mathrm{WCart}^{\mathrm{HT}}}$  to Liu’s cyclotomic chart has zero one-correspondence operator. After  $G = \mathbf{F}_p^\times$ -descent its two-term arithmetic complex is*

$$\left[ D_r \xrightarrow{0} D_r^{\mathrm{ar}} \right].$$

*Proof.* Write  $h = q(q-1)$ . On the universal cyclotomic chart,

$$\gamma_A(q) - q = q(q^p - 1) = dh.$$

Consequently

$$(2.8) \quad (\gamma_A - \mathrm{id})(A) \subset dhA, \quad \partial_A(A) = \frac{\gamma_A - \mathrm{id}}{h}(A) \subset dA.$$

Indeed, the first containment follows for polynomials from  $f(q+dh) - f(q) \in dhA$ , and then for  $A = \mathbf{Z}_p[[q-1]]$  by continuity, since the principal ideal  $dhA$  is closed in the Noetherian complete local ring  $A$ . Thus the scalar one-correspondence operator induced on  $R = A/(d)$  is exactly zero.

The adjacent Hodge quotient is  $D_r \widehat{\otimes} \mathcal{O}_{\mathrm{WCart}^{\mathrm{HT}}}$ . The bottom map in AKN Definition 6.6 uses the canonical trivialization of  $F^* \mathcal{O}\{i\}$  on the Hodge–Tate locus. For  $i = 1$ , this trivialization is the one obtained from the canonical identification of the Breuil–Kisin line modulo the Hodge–Tate ideal; arbitrary  $i$  follows by tensor powers. See [2, Remark 2.5.7, Example 3.5.2, and Construction 5.5.3]. Hence the quotient contains neither a residual Breuil–Kisin character nor a shift depending on  $i$ . The factor  $D_r$  is used here as an external constant coefficient. Liu’s target  $\mathrm{WCart}$  carries the trivial  $G$ -action, so its pullback has the corresponding trivial  $G$ -linearization. This concerns the linearization in the present descent diagram; it does not say that the relative de Rham complex  $D_r$ , equipped instead with its independent natural Galois action, is a trivial Galois representation. The external coefficient also has no cyclotomic one-correspondence operator. Therefore the operator on  $D_r \widehat{\otimes}_{\mathbf{Z}_p} R$  is  $1 \widehat{\otimes} \bar{\partial}_A = 0$ , not multiplication by a unit or by a Sen character.

Finally, Liu’s invariant calculation [3, Lemma 3.5.11] gives

$$R^G = \mathbf{Z}_p, \quad (R\epsilon_{\mathrm{ar}})^G = \mathbf{Z}_p v, \quad v = e\epsilon_{\mathrm{ar}}.$$

Since  $|G| = p - 1$  is invertible in  $\mathbf{Z}_p$ , invariants are exact and commute with tensoring by the trivially acted-on complex  $D_r$ . Taking invariants in the zero-operator complex therefore gives precisely  $[D_r \xrightarrow{0} D_r^{\mathrm{ar}}]$ . Notice that the arithmetic basis  $v = e\epsilon_{\mathrm{ar}}$  is chosen only after the zero operator has been computed on the Hodge quotient. Replacing  $\epsilon_{\mathrm{ar}}$  by  $v$  therefore cannot introduce a scalar operator or a character.  $\square$

**Lemma 2.6** (Liu descent preserves the adjacent cofiber). *There are natural equivalences  $\lambda_{r,i}^-$ ,  $\lambda_{r,i}$ , and  $\kappa_r$  which form a morphism of cofiber sequences*

$$(2.9) \quad \begin{array}{ccccc} P_{r,i}^- & \longrightarrow & P_{r,i} & \xrightarrow{\beta_{r,i}} & H_r \\ \lambda_{r,i}^- \downarrow \sim & & \lambda_{r,i} \downarrow \sim & & \kappa_r \downarrow \sim \\ [X_{r,i}^- \longrightarrow Y_{r,i}^-] & \longrightarrow & [X_{r,i} \xrightarrow{\partial_{r,i}^{\mathrm{ar}}} Y_{r,i}] & \longrightarrow & [D_r \xrightarrow{0} D_r^{\mathrm{ar}}]. \end{array}$$

Here the horizontal maps, including their nullhomotopies, are induced by the same functorial adjacent Nygaard cofiber sequence. Moreover,  $\kappa_r$  is an equivalence under  $D_r$ . Taking cofibers of the augmentations gives a second morphism of cofiber sequences

$$(2.10) \quad \begin{array}{ccccc} D_r & \longrightarrow & H_r = D_r \widehat{\otimes} \mathrm{R}\Gamma(\mathrm{WCart}^{\mathrm{HT}}, \mathcal{O}) & \longrightarrow & D_r[-1] \\ \parallel & & \kappa_r \downarrow \sim & & \bar{\kappa}_r \downarrow \sim \\ D_r & \longrightarrow & [D_r \xrightarrow{0} D_r^{\mathrm{ar}}] & \longrightarrow & D_r^{\mathrm{ar}}[-1]. \end{array}$$

*Proof.* Bhatt–Lurie’s functorial adjacent-piece sequence [2, Corollary 5.2.8 and Remark 5.5.2], including its Breuil–Kisin-twisted form from their Construction 5.5.3, identifies the cofiber of

$$I \otimes \mathrm{N}^{\geq r-1} F^* \mathcal{H}_i \longrightarrow \mathrm{N}^{\geq r} F^* \mathcal{H}_i$$

with

$$D_r \otimes \mathcal{O}_{\mathrm{WCart}^{\mathrm{HT}}}.$$

This is a cofiber sequence of quasi-coherent complexes, with a functorial specified nullhomotopy. Derived global sections are exact, so applying  $\mathrm{R}\Gamma(\mathrm{WCart}, -)$  gives the upper row of (2.9); its second arrow is exactly the Hodge–Tate comparison  $\beta_{r,i}$  used in the defining AKN square.

To make the required arrow-level naturality explicit, for  $\mathcal{F} \in \mathcal{D}(\mathrm{WCart})$  put

$$\mathcal{K}(\mathcal{F}) := \mathrm{fib} \left( (\rho^* \pi_R^* \mathcal{F})^G \xrightarrow{\partial_{\mathcal{F}}^{\mathrm{ar}}} (\rho^* \pi_R^* \mathcal{F} \otimes_A \mathcal{L}_{\mathrm{ar}})^G \right).$$

Liu's coefficient operator is natural in  $\mathcal{F}$  [3, Proposition 3.1.21]. Pullback, tensoring with  $\mathcal{L}_{\mathrm{ar}}$ , finite invariants, and fibers are exact; hence  $\mathcal{K}$  is an exact functor. Liu's global-sections equivalence [3, Proposition 3.3.1], followed by the finite descent of [3, Theorem 3.5.2], is a natural equivalence

$$\eta_{\mathcal{F}} : \mathrm{R}\Gamma(\mathrm{WCart}, \mathcal{F}) \xrightarrow{\sim} \mathcal{K}(\mathcal{F}).$$

Apply Lemma 2.4 to this natural equivalence and the functorial adjacent cofiber sequence above. Exactness of  $\mathcal{K}$  gives the lower row of (2.9), and naturality of  $\eta_{\mathcal{F}}$  gives the two commutative squares with their specified homotopy. The first two vertical maps are  $\lambda_{r,i}^-$  and  $\lambda_{r,i}$ . Since  $|G| = p - 1$  is invertible, homotopy fixed points agree with ordinary invariants and preserve the cofiber sequence and its specified nullhomotopy.

Lemma 2.5 identifies the quotient two-term complex, including the Tate trivialization, the scalar operator, the  $G$ -action, and the arithmetic line, as  $[D_r \xrightarrow{0} D_r^{\mathrm{ar}}]$ .

We now compare this complex with the Hodge–Tate corner without identifying their displayed integral bases. Apply Liu's Hodge–Tate one-correspondence fiber calculation [3, Proposition 3.2.3] together with the finite-descent equivalence [3, Theorem 3.5.3] to the coefficient  $D_r \hat{\otimes} \mathcal{O}_{\mathrm{WCart}^{\mathrm{HT}}}$ . The resulting natural map

$$\mathrm{R}\Gamma(\mathrm{WCart}^{\mathrm{HT}}, D_r \hat{\otimes} \mathcal{O}_{\mathrm{WCart}^{\mathrm{HT}}}) \longrightarrow \mathrm{R}\Gamma([R^{\mathrm{HT}}/G], D_r \hat{\otimes} \mathcal{O})$$

is an equivalence. Its source is  $H_r$ , while the arithmetic one-correspondence computes its target by  $[D_r \xrightarrow{0} D_r^{\mathrm{ar}}]$ . Pullback preserves the unit section. Naturality with respect to the quotient map in the adjacent sequence identifies this equivalence with the third vertical map  $\kappa_r$  in (2.9). Thus it is an equivalence under  $D_r$ , and exactness of cofibers gives (2.10). The homotopy in the right square is the image of the functorial nullhomotopy in Bhatt–Lurie's adjacent sequence.

The comparison used here is the augmented derived equivalence; it does not identify the two integral bases. Indeed, Liu's comparison between the integral  $q$ -Sen operator and the logarithmic Sen operator is nonlinear, although its reduction modulo  $p$  is termwise linear in the displayed coordinates.  $\square$

### 2.3.3. Identification of the logarithmic residue.

**Lemma 2.7** (Independence of the logarithmic coordinate). *The construction of  $\partial_{r,i}^{\mathrm{div}} : X_{r,i} \rightarrow Y_{r,i}^-$ , the complex  $\mathcal{C}_{r,i}$ , and the comparison map to that complex do not require an identification of the Bhatt–Lurie logarithm generator with  $v = e\epsilon_{\mathrm{ar}}$ . More precisely, replacing the displayed coordinate on the shifted summand in (2.10) by any automorphism of that invertible summand changes the vertical comparison equivalence but leaves the bottom adjacent square, its specified nullhomotopy, and its horizontal fiber unchanged.*

*Proof.* Before choosing a coordinate on either shifted line, (2.9) is a morphism of cofiber sequences in the enhanced stable category. Its bottom row and its specified 2-simplex are obtained by applying the exact functor  $\mathcal{K}$  of the proof of Lemma 2.6 to the adjacent AKN package. Consequently the lift

$$\partial_{r,i}^{\mathrm{div}} : X_{r,i} \longrightarrow Y_{r,i}^-$$

is determined by that 2-simplex before the cofiber  $D_r^{\mathrm{ar}}$  is trivialized.

To see explicitly what a coordinate change can do, let  $a : D_r^{\mathrm{ar}} \xrightarrow{\sim} D_r^{\mathrm{ar}}$  be an automorphism. Postcomposing the rightmost vertical equivalence in (2.10) by  $a[-1]$  replaces both sides of

$$\bar{\kappa}_r \mathrm{Res}_r \simeq \alpha_{r,i} \lambda_{r,i}$$



by their postcomposition with  $a[-1]$ . Intrinsically,  $\bar{\kappa}_r$  and the octahedral boundary are fixed: this operation only replaces the chosen identification of their common target with the named free line  $D_r^{\text{ar}}[-1]$ , so both coordinate expressions change together. Since  $a[-1]$  is an equivalence,  $\text{fib}(a[-1]\alpha_{r,i}) \rightarrow \text{fib}(\alpha_{r,i})$  is a canonical equivalence. The object  $Y_{r,i}^-$ , the lift through  $Y_{r,i}^- \rightarrow Y_{r,i}$ , and therefore the bottom horizontal fiber are unchanged. Thus the possible integral unit belongs only to a coordinate description of the comparison equivalence. It cannot alter  $\partial_{r,i}^{\text{div}}$  or any arrow in the  $q$ -complex, whose arithmetic bases were fixed independently by  $\epsilon_{\text{ar}} \mapsto d_{\Delta}q$  and  $\omega = d\epsilon_{\text{ar}}$ .  $\square$

*Remark 2.8* (Bypass certificate for the integral logarithm unit). The unknown integral unit between the Bhatt–Lurie logarithm basis and  $v = e\epsilon_{\text{ar}}$  is not a suppressed hypothesis. Its possible uses in the proof are exhausted by the following list.

- (1) The Bhatt–Lurie basis is used in (2.5) only to split the Hodge–Tate fiber as  $\mathbf{Z}_p \oplus \mathbf{Z}_p[-1]$ . The unit map is the first summand and is independent of a coordinate on the shifted summand.
- (2) In (2.9) and (2.10), the comparison with the Liu invariant line is used only as an equivalence under  $D_r$ . Lemma 2.7 shows that changing its shifted coordinate postcomposes an equivalence and gives a canonically equivalent fiber; it does not change the adjacent package map or its specified nullhomotopy.
- (3) The two horizontal  $q$ -operators are defined after this cancellation from Liu’s arithmetic coordinate  $\epsilon_{\text{ar}}$  and the scalar-one GLSQ identifications

$$\epsilon_{\text{ar}} \mapsto d_{\Delta}q, \quad \omega = d\epsilon_{\text{ar}} = d_{q^p}q.$$

Neither identification passes through the Bhatt–Lurie logarithm basis.

- (4) The associated-graded normalization uses only the oriented Rees generator  $d$  and  $u = dt^{-1}$ ; the two vertical syntomic maps use only the oriented Breuil–Kisin generator and  $\omega$ .

Thus every object and arrow on the right side of (1.2) is unchanged by the unknown unit. What varies is only a displayed coordinate for an intermediate comparison equivalence. Consequently no scalar-one Bhatt–Lurie/Liu logarithm comparison is needed for Theorem 1.1.

**Proposition 2.9** (Residue is the adjacent octahedral boundary). *Let*

$$\pi_{r,i}^{\text{ar}} : Y_{r,i} \longrightarrow D_r^{\text{ar}}$$

*be the cofiber arrow in (2.7). The specified nullhomotopy of  $\pi_{r,i}^{\text{ar}} \partial_{r,i}^{\text{ar}}$  determines a lift*

$$\partial_{r,i}^{\text{div}} : X_{r,i} \longrightarrow Y_{r,i}^-$$

*through  $u_{r,i}$ . Let*

$$\alpha_{r,i} : \text{fib}(\partial_{r,i}^{\text{ar}}) \longrightarrow \text{cofib}(u_{r,i})[-1] \simeq D_r^{\text{ar}}[-1]$$

*be the octahedral boundary of Lemma 2.3 for this specified lift. Then*

$$\bar{\kappa}_r \circ \text{Res}_r \simeq \alpha_{r,i} \circ \lambda_{r,i}.$$

*Consequently*

$$(2.11) \quad N_{r,i} \simeq \text{fib}\left(X_{r,i} \xrightarrow{\partial_{r,i}^{\text{div}}} Y_{r,i}^-\right).$$

*Proof.* Taking the two terms of Lemma 2.6 gives (2.7); taking its differential gives a specified nullhomotopy of  $X_{r,i} \rightarrow Y_{r,i} \rightarrow D_r^{\text{ar}}$ . Hence the fiber description  $Y_{r,i}^- \simeq \text{fib}(\pi_{r,i}^{\text{ar}})$  identifies this nullhomotopy with the stated lift  $\partial_{r,i}^{\text{div}}$ . Thus the factorization is canonical relative to the specified nullhomotopy.

By definition of  $\text{Res}_r$ , the upper composite below is  $\text{Res}_r$ . Lemma 2.6 gives the entire commutative diagram

$$(2.12) \quad \begin{array}{ccccc} P_{r,i} & \xrightarrow{\beta_{r,i}} & H_r & \longrightarrow & D_r[-1] \\ \lambda_{r,i} \downarrow \sim & & \kappa_r \downarrow \sim & & \bar{\kappa}_r \downarrow \sim \\ [X_{r,i} \xrightarrow{\partial_{r,i}^{\text{ar}}} Y_{r,i}] & \longrightarrow & [D_r \xrightarrow{0} D_r^{\text{ar}}] & \longrightarrow & D_r^{\text{ar}}[-1]. \end{array}$$

The left square is the middle-right face of (2.9); the right square is obtained from the two augmentations by the functorial cofiber construction (2.10). Thus the diagram also remembers the specified nullhomotopy.

For completeness, write the map of arrows underlying the bottom-left map as

$$\begin{array}{ccc} X_{r,i} & \xrightarrow{\partial_{r,i}^{\text{ar}}} & Y_{r,i} \\ \pi_{r,i}^0 \downarrow & & \downarrow \pi_{r,i}^{\text{ar}} \\ D_r & \xrightarrow{0} & D_r^{\text{ar}}. \end{array}$$

This square includes the specified homotopy  $\pi_{r,i}^{\text{ar}} \partial_{r,i}^{\text{ar}} \simeq 0$ . This homotopy is precisely the extra datum needed to lift  $\pi_{r,i}^{\text{ar}}$  through the mapping-space fiber

$$\begin{aligned} & \text{Map}(\text{cofib}(\partial_{r,i}^{\text{ar}}), D_r^{\text{ar}}) \\ & \simeq \text{fib}(\text{Map}(Y_{r,i}, D_r^{\text{ar}}) \longrightarrow \text{Map}(X_{r,i}, D_r^{\text{ar}})). \end{aligned}$$

It therefore determines a canonical morphism

$$\tilde{\pi}_{r,i}^{\text{ar}} : \text{fib}(\partial_{r,i}^{\text{ar}}) = \text{cofib}(\partial_{r,i}^{\text{ar}})[-1] \longrightarrow D_r^{\text{ar}}[-1].$$

Passing to fibers and then taking the cofiber of the augmentation by  $D_r$  is exactly this construction: the unshifted  $D_r$ -component is removed, while  $\tilde{\pi}_{r,i}^{\text{ar}}$  remains. In an ordinary two-term cone model it is represented by  $(0, \pi_{r,i}^{\text{ar}})$ , with the displayed homotopy supplying the chain-map homotopy. With the fixed convention  $\text{fib}(f) = \text{Cone}(f)[-1]$ , the cone model for the octahedral sequence of  $X_{r,i} \rightarrow Y_{r,i}^- \rightarrow Y_{r,i}$  identifies this map with  $\alpha_{r,i}$ ; it is not a projection from a general fiber to the shifted target. Hence (2.12) identifies this map with  $\bar{\kappa}_r \circ \text{Res}_r$ , including the homotopy which gives the canonical factorization through  $Y_{r,i}^-$ . Thus Lemma 2.3 applies to the displayed factorization. Since  $\bar{\kappa}_r$  is an equivalence, Lemma 2.2 and octahedral cancellation give (2.11).  $\square$

*Remark 2.10.* The full complex  $D_r$  is essential in Proposition 2.9. Only after the residue has been identified with the adjacent octahedral boundary may the Hodge-completed term be cancelled. In particular, one cannot replace  $D_r$  by  $R$  before applying the adjacent cofiber theorem.

**2.4. The filtered  $q$ -model.** Under the oriented generator of the Breuil–Kisin line, the calculation of [4, Lemma 2.15, Remark 2.16(1), and Example 3 following Proposition 2.19] identifies its arithmetic  $\gamma$ -action with multiplication by  $\lambda(q)^i$ ; after Frobenius pullback the factor is  $\lambda(q^p)^i$ . Thus the operators in (1.1) are the two arithmetic actions attached to the Tate line.

**Lemma 2.11** (Integral operators and horizontality). *The operators  $D_i$  and  $\tilde{D}_i$  preserve  $A$ . Moreover, for every  $a \in A$  and  $r \geq 1$ ,*

$$(2.13) \quad \tilde{D}_i(d^r a) = d^{r-1} \frac{g^{r-i} r_i(a) - a}{h},$$

$$(2.14) \quad \tilde{D}_i(\varphi(a)) = q^{p-1} \varphi(D_i(a)).$$

For  $r = 0$ , one has  $g^{-i}r_i = \tilde{r}_i$  and  $(\tilde{r}_i - 1)/h = d\tilde{D}_i$ .

*Proof.* Both numerators vanish at  $q = 1$ , so they are divisible by  $q - 1$ . Modulo  $d$  one has  $q^p = 1$ , hence  $q^{p+1} = q$  and  $[p + 1]_{q^p} = p + 1$ ; therefore  $\tilde{r}_i \equiv \text{id} \pmod{d}$ . The elements  $d$  and  $q - 1$  are relatively prime in the UFD  $A$ : the element  $q - 1$  is prime and does not divide  $d$ , since  $d|_{q=1} = p \neq 0$ . Hence  $(d) \cap (q - 1) = (d(q - 1))$ . Since  $q$  is a unit, this proves the integrality of  $\tilde{D}_i$ ; the proof for  $D_i$  is the first divisibility alone.

Writing both sides as quotients of  $q^{p(p+1)} - 1$  gives

$$g = \frac{\gamma(d)}{d} = \frac{[p + 1]_{q^p}}{[p + 1]_q}, \quad \lambda(q^p)g = \lambda(q).$$

Using  $\gamma(d) = dg$  and  $\lambda(q^p) = \lambda(q)g^{-1}$ , one computes

$$\begin{aligned} \tilde{r}_i(d^r a) &= \lambda(q^p)^i \gamma(d)^r \gamma(a) \\ &= d^r g^{r-i} \lambda(q)^i \gamma(a) = d^r g^{r-i} r_i(a). \end{aligned}$$

Subtracting  $d^r a$  and dividing by  $dh$  proves (2.13), including the stated  $r = 0$  formula. Finally,  $\tilde{r}_i \varphi = \varphi r_i$  and  $\varphi(h) = q^{p-1}dh$ ; division gives (2.14).  $\square$

Let  $\epsilon_{\text{ar}}$  be Liu's arithmetic differential basis, identified with  $d_{\Delta}q$  by the normalization fixed before Theorem 1.1, and put

$$\omega = d\epsilon_{\text{ar}} = d_{q^p}q.$$

For  $r \geq 0$ , define a two-term complex

$$(2.15) \quad \mathcal{C}_{r,i} = \left[ d^r A \xrightarrow{\nabla_{r,i}} L_r \right], \quad L_r = \begin{cases} A\omega, & r = 0, \\ d^{r-1}A\omega, & r \geq 1. \end{cases}$$

The differential is the restriction of  $\tilde{D}_i$ , namely

$$\nabla_{r,i}(d^r a) = \tilde{D}_i(d^r a)\omega.$$

For  $r \geq 1$ , put

$$\rho_{r|i} = g^{r-i}r_i, \quad \mathcal{Q}_{r|i} = \frac{\rho_{r|i} - 1}{h}.$$

Then

$$\tilde{D}_i(d^r a) = d^{r-1}\mathcal{Q}_{r|i}(a).$$

For  $r = 0$ , one has

$$\rho_{0|i} = \tilde{r}_i, \quad \frac{\rho_{0|i} - 1}{h} = d\tilde{D}_i.$$

Thus the undivided arithmetic operator lands in  $dA\epsilon_{\text{ar}} = A\omega$ ; its factorization through the adjacent term is precisely the bottom differential in (2.15). This explains why the degree-one terms at levels zero and one are equal. Replacing the level-zero term by  $A\epsilon_{\text{ar}}$  would instead introduce an additional Hodge quotient which is absent from the Nygaard pullback.

The inclusions  $\mathcal{C}_{r+1,i} \rightarrow \mathcal{C}_{r,i}$  make  $\mathcal{C}_{\bullet,i}$  a decreasing complete filtered complex. Its graded pieces are

$$(2.16) \quad \text{gr}^0 \mathcal{C}_{\bullet,i} \simeq R[0],$$

$$(2.17) \quad \text{gr}^r \mathcal{C}_{\bullet,i} \simeq \left[ R \xrightarrow{e[r]_s} R \right] \quad (r \geq 1),$$

where

$$[r]_s = \frac{s^r - 1}{s - 1}, \quad eh = p \text{ in } R.$$

Indeed, modulo  $d$ , one has

$$\gamma = \text{id}_R, \quad g = s, \quad r_i = s^i, \quad e q(q-1) = p.$$

Thus the differential on the  $r$ -th graded piece, for  $r \geq 1$ , is multiplication by

$$\frac{s^r - 1}{h} = e \frac{s^r - 1}{p} = e[r]_s.$$

At  $r = 0$ , the two degree-one lattices agree, so the degree-one graded term is zero and  $\text{gr}^0 \mathcal{C}_{\bullet, i} = R[0]$ .

**Lemma 2.12** (Evaluation of the adjacent Liu square). *For every  $r \geq 0$ , pullback from  $\text{WCart}$  to  $R^\Delta$ , followed by multiplication along the diagonal, induces a commutative square*

$$(2.18) \quad \begin{array}{ccc} X_{r,i} & \xrightarrow{\partial_{r,i}^{\text{div}}} & Y_{r,i}^- \\ \text{ev}_{r,i}^0 \downarrow & & \downarrow \text{ev}_{r,i}^1 \\ d^r A & \xrightarrow{\nabla_{r,i}} & L_r. \end{array}$$

At  $r = 0$ , the right vertical map factors through  $L_0 = dA\epsilon_{\text{ar}} = A\omega \subset A\epsilon_{\text{ar}}$ . As  $r$  varies, these squares are the Nygaard components of a morphism of complete AKN packages. In particular, they commute with the filtration transition maps, and the comparison from level  $i$  to level 0 commutes separately with  $\text{can}$  and with  $c_{R/A;0,i} \circ \varphi_{R/A;i,i}$ .

*Proof.* On  $R^\Delta$ , put

$$\mathcal{I}_{r,i} := \mathcal{I}^{r-i} \mathcal{O}\{i\}.$$

Here  $\mathcal{I}$  is an invertible Cartier-divisor line. For every integer  $m$ , including  $m < 0$ , the notation  $\mathcal{I}^m$  means its tensor power, with negative powers defined by tensor powers of  $\mathcal{I}^\vee$ . Thus  $\mathcal{I}_{r,i}$  is a genuine invertible coefficient for all  $r \geq 0$ ; no fractional ideal is being introduced when  $r < i$ . Let  $(B, I_B)$  be an oriented transversal prism and let  $x : R \rightarrow \overline{B} = B/I_B$ . Set

$$T_x = R \widehat{\otimes}_{\mathbf{Z}_p} \overline{B}, \quad m_x : T_x \longrightarrow \overline{B}, \quad a \otimes b \longmapsto x(a)b.$$

The structure map  $B \rightarrow T_x$  is  $b \mapsto 1 \otimes \bar{b}$ , and its composite with  $m_x$  is the quotient map  $B \rightarrow \overline{B}$ . Since  $R$  is finite free over  $\mathbf{Z}_p$ , the completed tensor product  $T_x$  is finite free as an  $\overline{B}$ -module. In particular,  $T_x$  is derived  $p$ -complete and has bounded  $p$ -power torsion whenever  $\overline{B}$  does. Thus both  $(B, T_x)$  and  $(B, \overline{B})$  lie in the category of bounded  $\delta$ -pairs used by AKN, and the displayed maps form a morphism in that category. The map used below must be distinguished from the comparison map of AKN Definition 6.12. That definition compares, for one fixed prismatic  $\delta$ -pair, AKN's globally defined Nygaard filtration with the usual relative Nygaard filtration; AKN Proposition 6.13 says that this comparison is an equivalence. It does *not* say that the morphism induced by  $m_x$  between the two different pairs below is an equivalence. We use only the ordinary functorial map induced by  $m_x$ . We use the ring-map convention for relative prismatic cohomology: a  $B/I_B$ -algebra map  $S \rightarrow S'$  induces a map  $\Delta_{S/B} \rightarrow \Delta_{S'/B}$ . Thus  $m_x$  induces a map from the relative prismatic complex of  $T_x$  over  $B$  to that of  $\overline{B}$  over  $B$ . The latter is the oriented base prism.

More is true and will be needed for the syntomic map. The ring map  $m_x$  is a morphism of bounded  $\delta$ -pairs

$$(B, T_x) \longrightarrow (B, \overline{B}).$$

Write

$$\mathcal{P}_{S/B}^{\text{AKN}} := (H_{S/B}, N_{S/B}, c_{S/B}, \varphi_{S/B})$$

for the full AKN relative prismatic package. Here  $c : H \rightarrow \text{sep}(N)$  has the Day-convolution target of AKN Construction 7.3 and  $\varphi : N \rightarrow \text{sh}(H)$ ; neither target will be replaced by a naive single tensor product. Theorem 1.2 of AKN constructs this package functorially in bounded  $\delta$ -pairs, and AKN

Construction 10.44 states explicitly that the two syntomic structure maps  $\text{can}$  and  $c\varphi$  are functorial. Consequently [1, Theorem 1.2 and Construction 10.44] gives a morphism of packages

$$(2.19) \quad \mathcal{P}_{T_x/B}^{\text{AKN}} \longrightarrow \mathcal{P}_{\overline{B}/B}^{\text{AKN}}.$$

At the only component needed here, where filtration degree and Tate degree are both  $i$ , the same morphism contains the square

$$\begin{array}{ccc} N^{\geq i} \widehat{\Delta}_{T_x/B}^{(1)} \{i\} & \xrightarrow{c_{T_x/B;0,i} \varphi_{T_x/B;i,i}} & \widehat{\Delta}_{T_x/B}^{(1)} \{i\} \\ \downarrow & & \downarrow \\ N^{\geq i} \widehat{\Delta}_{\overline{B}/B}^{(1)} \{i\} & \xrightarrow{c_{\overline{B}/B;0,i} \varphi_{\overline{B}/B;i,i}} & \widehat{\Delta}_{\overline{B}/B}^{(1)} \{i\}. \end{array}$$

The targets are the level-zero Nygaard-completed terms. The corresponding square for  $\text{can}$  is obtained by replacing both horizontal arrows by the filtration inclusions. This is the precise arrow-level compatibility needed for the syntomic fiber; no assertion about every individual component of  $c$  is required.

We spell out the target of its Nygaard component, since this is where the shift by  $i$  enters. By the relative-prismatic comparison in [1, Theorem 1.2(2)], the package for the target pair  $(B, \overline{B})$  is the usual relative package of the base prism. Bhatt–Lurie [2, Example 5.1.4] identifies its untwisted level- $r$  Nygaard term with  $I_B^r$  for  $r \geq 0$ . On the other hand, [2, Remark 2.5.9 and Remark 3.6.5] gives

$$F^* \mathcal{O}\{i\} \simeq I_B^{-i} \mathcal{O}\{i\}.$$

Consequently the level- $r$ ,  $i$ -twisted target of (2.19) is exactly

$$I_B^r \otimes_B I_B^{-i} B\{i\} = I_B^{r-i} B\{i\},$$

including when  $r < i$ , since  $I_B$  is invertible. The construction is functorial under morphisms of oriented transversal prisms and under base change of the point  $x$ . Hence these pointwise maps satisfy the crystal compatibilities and descend to a morphism

$$(2.20) \quad \pi_R^* \mathcal{E}_{r,i} \longrightarrow \mathcal{J}_{r,i}$$

on  $R^\Delta$ . It commutes with multiplication by  $I_B$ . Independently, the adjacent Nygaard fiber sequence and its nullhomotopy are functorial in the relative algebra by Bhatt–Lurie’s Corollary 5.2.8, Remark 5.5.2, and Construction 5.5.3. Hence the same map commutes with the adjacent Nygaard arrows and with their specified nullhomotopies. AKN Definition 6.12 and Proposition 6.13 enter only to identify AKN’s global construction with the usual relative Nygaard package at a fixed prismatic pair; they do not turn (2.20) into an equivalence.

Pull back to Liu’s cyclotomic chart. His one-correspondence construction is functorial in morphisms of coefficients, so the preceding map gives a map of the two-term arithmetic complexes and of their adjacent cofiber sequences. The source terms in (2.18) were written using finite descent. To form the displayed vertical maps, put  $J_{r,i}^{\text{ch}} := \rho^* \mathcal{J}_{r,i}$  and use the explicit factorization

$$M_{r,i}^{hG} \xrightarrow{(\text{ev}_{m_x})^{hG}} (J_{r,i}^{\text{ch}})^{hG} \xrightarrow{\text{res}} J_{r,i}^{\text{ch}}.$$

Since  $|G|$  is invertible, homotopy fixed points agree with the ordinary invariants used in the notation  $X_{r,i} = M_{r,i}^G$ . The target is the full  $A$ -module on the chosen chart. At this stage the vertical maps are maps of  $\mathbf{Z}_p$ -complexes; after passage to the central associated graded, the required  $R$ -linearity follows from Proposition 2.14.

Write  $e_i$  for the oriented Breuil–Kisin generator. We use the coordinate

$$d^{r-i} e_i \longmapsto d^r$$

on  $\mathcal{J}_{r,i} = \mathcal{I}^{r-i}\mathcal{O}\{i\}$ . This is an  $A$ -linear coordinate; its nontrivial  $\gamma$ -change is exactly the factor  $g^{r-i}r_i$  below. For  $r \geq 1$ , the evaluated filtration generator is therefore  $d^r$ , while the evaluated adjacent arithmetic line is

$$d^r A\epsilon_{\text{ar}} = d^{r-1} A\omega = L_r.$$

The arithmetic action on this generator is  $\rho_{r|i} = g^{r-i}r_i$ . Hence functoriality of Liu's operator identifies the bottom differential with

$$d^r a \mapsto d^r \frac{\rho_{r|i}(a) - a}{h} \epsilon_{\text{ar}} = d^{r-1} \mathcal{Q}_{r|i}(a)\omega = \tilde{D}_i(d^r a)\omega.$$

For  $r = 0$ , exhaustivity identifies  $\mathcal{E}_{-1,i} \simeq \mathcal{E}_{0,i}$ , but the adjacent map is still the morphism  $\mathcal{I} \otimes \mathcal{E}_{-1,i} \rightarrow \mathcal{E}_{0,i}$ . The evaluation used on the first term is therefore the level-zero map  $\mathcal{E}_{0,i} \rightarrow \mathcal{J}_{0,i}$ . After tensoring it with  $\mathcal{I}$ , its target is  $\mathcal{I}\mathcal{J}_{0,i} = \mathcal{J}_{1,i}$ , rather than a formally introduced negative filtration level. On the oriented chart this is multiplication by  $d$ . Therefore the evaluated right-hand term is

$$dA\epsilon_{\text{ar}} = A\omega = L_0.$$

Moreover  $\rho_{0|i} = \tilde{r}_i$ , so  $(\rho_{0|i} - 1)/h = d\tilde{D}_i$ ; the same calculation gives the differential  $\nabla_{0,i}$ . This proves the square, including the asserted level-zero divisibility.  $\square$

**Proposition 2.13** (Diagonal-retraction comparison). *There is a canonical morphism of complete filtered complexes*

$$\theta_{\bullet,i} : (N_{\bullet,i})^\wedge \longrightarrow \mathcal{C}_{\bullet,i}.$$

*It is induced by a  $G$ -equivariant natural transformation of filtered prismatic crystals. Hence its level maps commute with the adjacent filtration arrows, and after completion the square*

$$\begin{array}{ccc} (N_{i,i})^\wedge & \xrightarrow{c\varphi} & (N_{0,i})^\wedge \\ \theta_{i,i} \downarrow & & \downarrow \theta_{0,i} \\ \mathcal{C}_{i,i} & \xrightarrow{\Phi_C} & \mathcal{C}_{0,i} \end{array}$$

*commutes, where  $\Phi_C$  is the map induced by the bottom package's composite  $c\varphi$  on the evaluated complexes.*

*Proof.* By Proposition 2.9, the true level- $r$  object is the fiber of the upper arrow in (2.18); by definition, the filtered  $\mathcal{C}_{r,i}$  is the fiber of its lower arrow. Taking fibers in that square gives a morphism  $N_{r,i} \rightarrow \mathcal{C}_{r,i}$ .

Equivalently, before passing to the divided-fiber presentation, the pointwise construction is a map

$$\pi_R^* \mathcal{E}_{r,i} \longrightarrow \mathcal{J}_{r,i}$$

on  $R^\Delta$ , where  $\mathcal{E}_{r,i}$  is the pointwise Nygaard crystal defined before (2.6), and  $\mathcal{J}_{r,i} = \mathcal{I}^{r-i}\mathcal{O}\{i\}$ . Under the adjunction  $\pi_R^* \dashv \pi_{R,*}$ , it has the correct direction  $\mathcal{E}_{r,i} \rightarrow \pi_{R,*} \mathcal{J}_{r,i}$ . At level zero the adjacent square of Lemma 2.12, rather than this adjunction alone, is what factors the degree-one component through  $dA\epsilon_{\text{ar}}$ . The opposite structural unit induced by  $s_x : \bar{B} \rightarrow T_x$  only normalizes the multiplication retraction  $m_x$ ; it cannot produce a comparison in the opposite direction.

For a morphism of transversal prisms, the multiplication maps  $m_x$  form a commutative square of algebra maps. The ring-map functoriality stated in the proof of Lemma 2.12 makes (2.20) a natural transformation of filtered crystals. It is  $G$ -equivariant because the multiplication maps are equivariant for Liu's cyclotomic action. Thus it descends through finite homotopy fixed points and assembles in  $r$ .

Compatibility with  $c\varphi$  is not deduced from the universal property of completion. Pointwise it is the syntomic-component square of the morphism of AKN packages (2.19), displayed in the proof of Lemma 2.12. Pullback to Liu's chart, the one-correspondence functor, finite homotopy fixed points,

fibers, and filtered completion are exact functors, so they preserve this commutative diagram. The resulting completed square is exactly the square in the statement.  $\square$

## 2.5. Central fiber and completion.

2.5.1. *The central Weyl model.* The comparison map is now tested on the associated graded of the Nygaard filtration. The level-zero calculation is separated from the positive levels because the two degree-one lattices agree at that endpoint.

**Proposition 2.14** (Derived central Weyl dictionary). *Put  $R_n = R/p^n$ , and let*

$$\mathcal{X}_n = \left[ \mathbf{A}_{-,R_n}^1 / (\mathbf{G}_a^\# \rtimes \mathbf{G}_m)_{R_n} \right].$$

Let

$$\mathcal{W}_{R,n} = R_n \langle u, D \rangle / (Du - uD - e), \quad \deg(u) = -1, \quad \deg(D) = 1.$$

Denote by  $\mathcal{D}_{\text{gr},D\text{-tors}}(\mathcal{W}_{R,n})$  the full stable subcategory of the derived category of graded left  $\mathcal{W}_{R,n}$ -modules whose cohomology is locally nilpotent under  $D$ : for every homogeneous class  $x$ , there is an integer  $m = m(x)$  such that  $D^m x = 0$ . There is an equivalence of stable categories

$$(2.21) \quad \mathcal{D}_{\text{qc}}(\mathcal{X}_n) \simeq \mathcal{D}_{\text{gr},D\text{-tors}}(\mathcal{W}_{R,n}).$$

Under this equivalence the structure sheaf corresponds to  $R_n[u]$ , with  $D = e d/du$ .

Let

$$\mathcal{X} = \left[ \mathbf{A}_{-,R}^1 / (\mathbf{G}_a^\# \rtimes \mathbf{G}_m)_R \right] \simeq (R^{\text{Nyg}})_{t=0}.$$

Here  $\mathcal{D}_{\text{qc}}(\mathcal{X})_p^\wedge$  denotes the full stable subcategory of quasi-coherent complexes whose pullbacks to the affine quotient nerve are derived  $p$ -complete. Passing to derived  $p$ -completion gives

$$(2.22) \quad \mathcal{D}_{\text{qc}}(\mathcal{X})_p^\wedge \simeq \varprojlim_n \mathcal{D}_{\text{gr},D\text{-tors}}(\mathcal{W}_{R,n}).$$

Consequently, if  $E, F$  correspond to systems  $(M_n)$  and  $(N_n)$  equipped with transition equivalences

$$M_{n+1} \otimes_{R_{n+1}}^{\mathbf{L}} R_n \simeq M_n, \quad N_{n+1} \otimes_{R_{n+1}}^{\mathbf{L}} R_n \simeq N_n,$$

then

$$(2.23) \quad \text{RHom}_{\mathcal{X}}(E, F) \simeq R \varprojlim_n \text{RHom}_{\mathcal{W}_{R,n}}^{\text{gr}}(M_n, N_n).$$

In particular, the  $p$ -complete structure sheaf corresponds to

$$\widehat{R[u]} = \varprojlim_n R_n[u].$$

*Proof. Finite-level Cartier duality.* We first work modulo  $p^n$ . Write

$$\mathcal{C}_n = \mathcal{D}_{\text{qc}}(B(\mathbf{G}_a^\#)_{R_n}).$$

Bhatt's derived Cartier-duality theorem and its relative form [6, Propositions 2.4.4–2.4.5] give a symmetric monoidal equivalence

$$\mathcal{C}_n \simeq \mathcal{D}_{(D)\text{-comp}}(R_n[D]),$$

where the right-hand side is the category of quasi-coherent complexes on the formal dual line  $\widehat{\mathbf{G}}_{a,R_n}$ , and its monoidal structure is convolution. Under this equivalence, pullback along the tautological point is local cohomology at  $D = 0$ . Composing with the complete-torsion equivalence recalled in [6, Proposition 2.4.4, footnote 21] therefore gives an equivalence

$$(2.24) \quad \Psi_n : \mathcal{C}_n \xrightarrow{\sim} \mathcal{D}_{(D)\text{-tors}}(R_n[D]).$$

The category on the right is the full stable subcategory of  $\mathcal{D}(R_n[D])$  consisting of complexes on whose cohomology every element is killed by some power of  $D$ . This is the required derived statement; no passage through an equivalence of abelian hearts is being used.

*Monoidal transport and the Weyl module category.* We spell out the monoidal structure, since it is not enough merely to compose Cartier duality with complete–torsion duality and assert that the latter preserves ordinary tensor products. By [6, Proposition 2.4.4(1)], the underlying  $R_n$ -complex of  $\Psi_n(M)$  is the pullback of the representation  $M$  to the tautological point. On a representation in the heart, if  $a$  denotes the divided-power coordinate of  $\mathbf{G}_a^\sharp$ , the corresponding coaction has the form

$$m \mapsto \sum_{j \geq 0} D_M^j(m) a^{[j]}.$$

The tensor-product coaction is the product of the two coactions. Its coefficient of  $a^{[1]}$  is therefore

$$D_{M \otimes N} = D_M \otimes 1 + 1 \otimes D_N.$$

This is the calculation of the primitive element in the Hopf algebra defining the group action. Both sides of the asserted monoidal comparison are exact and preserve colimits in each variable, while the derived torsion category is generated under shifts, cones, and colimits by its heart (equivalently by the modules  $R_n[D]/D^j$ ). The heart calculation therefore determines the derived comparison. It identifies the convolution compatibility of [6, Proposition 2.4.4(2)], after the complete–torsion realization, with ordinary  $R_n$ -tensor product equipped with the displayed diagonal operator. In particular, we are not using an unproved assertion that complete–torsion duality is symmetric monoidal for ordinary tensor products.

Now use the affine equivariant dictionary before imposing the  $\mathbf{G}_m$ -action. Quasi-coherent complexes on  $[\mathbf{A}_{-,R_n}^1/\mathbf{G}_a^\sharp]$  are modules in  $\mathcal{C}_n$  over the commutative algebra object  $R_n[u]$  whose translation coaction is

$$u \mapsto u + ea.$$

Under  $\Psi_n$ , this algebra object is  $R_n[u]$  with  $D = e d/du$ . A module over it is therefore an  $R_n[u]$ -complex  $M$  with a  $D$ -action such that the action map

$$R_n[u] \otimes_{R_n} M \longrightarrow M$$

is  $D$ -linear for the diagonal operator on the source. Evaluating this condition on  $u \otimes m$  gives

$$D(um) = uD(m) + e m,$$

so these module objects are left modules over  $\mathscr{W}_{R,n}$  whose underlying  $R_n[D]$ -complex is  $D$ -power torsion. This gives an equivalence of derived module categories. Indeed, after forgetting the algebra action, the monad for modules over the displayed algebra object is

$$M \mapsto R_n[u] \otimes_{R_n} M$$

with diagonal operator  $D(f \otimes m) = e f' \otimes m + f \otimes Dm$ . An algebra over this monad is exactly a derived  $R_n[u]$ -module with an endomorphism satisfying  $Du - uD = e$ , hence exactly a derived left  $\mathscr{W}_{R,n}$ -module. No additional divided-power differential operator is missing: the divided-power integrability is precisely the condition that the restricted  $R_n[D]$ -complex have  $D$ -power-torsion cohomology, supplied by Bhatt’s Cartier-duality theorem.

The transported monoidal structure just identified shows, in particular, that  $\mathcal{D}_{(D)\text{-tors}}(R_n[D])$  is a full monoidal subcategory. The induced category of  $R_n[u]$ -modules is likewise full in the ambient derived category of  $\mathscr{W}_{R,n}$ -modules. Finally,  $\mathbf{G}_m$ -equivariance for the semidirect product records the grading  $\deg(u) = -1$ ,  $\deg(D) = 1$ . This proves (2.21), including its assertion about mapping spectra. The coordinate calculation agrees with the abelian calculation in [5, Lemma 3.11]; only Bhatt’s theorem and the complete–torsion equivalence supply the derived enhancement.

The algebra object itself, viewed as a module over itself, corresponds to the left structure module  $R_n[u]$  with  $D = e d/du$ . This identifies the unit object.



*Passage to derived  $p$ -completion.* It remains to pass to the formal quotient. Let  $B^\bullet$  be the affine bar nerve of  $\mathbf{A}_{-,R}^1 \rightarrow \mathcal{X}$ , and let  $B_n^\bullet = B^\bullet \otimes_R^L R_n$ . Quasi-coherent descent, the affine identity  $\mathcal{D}(B^j)_p^\wedge \simeq \varprojlim_n \mathcal{D}(B_n^j)$ , and commutation of limits give

$$\mathcal{D}_{\text{qc}}(\mathcal{X})_p^\wedge \simeq \text{Tot}_{[j] \in \Delta} \mathcal{D}(B^j)_p^\wedge \simeq \varprojlim_n \text{Tot}_{[j] \in \Delta} \mathcal{D}(B_n^j) \simeq \varprojlim_n \mathcal{D}_{\text{qc}}(\mathcal{X}_n).$$

For completeness, the affine identity used here is the derived  $p$ -completion statement, not an underived limit assertion. For a connective affine ring  $C$ , the two functors

$$M \mapsto (M \otimes_C^L C/p^n)_n, \quad (M_n)_n \mapsto R \varprojlim_n M_n$$

are inverse between derived  $p$ -complete  $C$ -complexes and compatible systems of  $C/p^n$ -complexes. The first composite is the defining completion equivalence  $M \simeq R \varprojlim_n (M \otimes_C^L C/p^n)$ ; for the second, the derived quotient  $C/p^n = \text{cofib}(p^n : C \rightarrow C)$  is perfect, so tensoring with it commutes with inverse limits. Base change modulo  $p^n$  and the transition equivalences of the system give  $(R \varprojlim_m M_m) \otimes_C^L C/p^n \simeq M_n$ . Applying this degree-wise to the affine bar nerve proves the stated affine identity. Limits commute with the totalization because both are limits in the same ambient category. The finite-level equivalences above commute with reduction in  $n$ , so taking their inverse limit proves (2.22). Mapping objects in a limit of stable categories are limits of the finite-level mapping objects; fullness at every finite level therefore gives (2.23). The compatible unit modules have inverse limit  $\widehat{R[u]}$ , as claimed.  $\square$

**Lemma 2.15** (No central shift from the Tate index). *For fixed Nygaard degree  $r$ , the central Rees weight is  $-r$ , independently of the Tate index  $i$ . After AKN's shearing, the central coefficient of  $F^*\mathcal{O}\{i\}$  is therefore the structure module  $R[u]$ , with no additional graded shift or character.*

*Proof.* There are two a priori different weights. The integer  $i$  is the Breuil–Kisin grading, whereas the  $\mathbf{G}_m$ -weight on the central Rees fiber records the Nygaard filtration degree. They cancel in the coefficient before the latter weight is read. More precisely, Bhatt–Lurie's identification

$$F^*\mathcal{O}\{i\} \simeq \mathcal{I}^{-i} \otimes \mathcal{O}\{i\}$$

on the Hodge–Tate locus is the trivialization used in AKN Definition 6.6. The Sen operator is 1 on  $\mathcal{I}$  and  $i$  on  $\mathcal{O}\{i\}$ ; by the Leibniz rule it is therefore

$$-i + i = 0$$

on  $F^*\mathcal{O}\{i\}$ . Thus this coefficient has the trivial Hodge–Tate character.

The same cancellation is visible intrinsically in AKN Lemma 6.10. For an arbitrary twist index  $j$ , its top fiber sequence is

$$\text{gr}_N^r \widehat{\Delta}_{R/\mathbf{Z}_p}^{(1)} \{j\} \longrightarrow F_{\leq r}^{\text{conj}} \Omega_{R/\mathbf{Z}_p}^{\text{diff}} \xrightarrow{\Theta+r} F_{\leq r-1}^{\text{conj}} \Omega_{R/\mathbf{Z}_p}^{\text{diff}}.$$

The right-hand terms and the operator involve  $r$ , but not  $j$ : the twist has already disappeared under the preceding Hodge–Tate trivialization and AKN's shearing. Finally, Bhatt's Rees convention  $\text{gr}^r F = R\Gamma(B\mathbf{G}_m, i^* \text{Rees}(F)(-r))$  identifies filtration degree  $r$  with central weight  $-r$ . Hence no  $i$ -dependent shift remains.  $\square$

**Lemma 2.16** (Central global sections). *For  $n \geq 1$ , let*

$$\mathcal{W}_{R,n} = (R/p^n)\langle u, D \rangle / (Du - uD - e), \quad \deg(u) = -1, \quad \deg(D) = 1.$$

*Put*

$$\widehat{\mathcal{W}}_R = \varprojlim_n \mathcal{W}_{R,n}, \quad \widehat{R[u]} = \varprojlim_n (R/p^n)[u].$$

Thus  $\widehat{R[u]}$  is the  $p$ -adic completion of  $R[u]$ : its elements are sums  $\sum_{j \geq 0} a_j u^j$  with  $a_j \rightarrow 0$   $p$ -adically. It is a restricted-series ring, not the ordinary power series ring  $R[[u]]$ . Normalize character twists so that  $\mathcal{O}_{\mathcal{X}}(1)$  corresponds to a generator in graded degree  $-1$ . Then, for every  $r \geq 0$ ,

$$R\Gamma(\mathcal{X}, \mathcal{O}_{\mathcal{X}}(-r)) \simeq \left[ Ru^r \xrightarrow{D} Ru^{r-1} \right],$$

where  $D(u^r) = r e u^{r-1}$  and the second term is zero for  $r = 0$ .

*Proof.* Proposition 2.14 supplies the derived central-stack dictionary and identifies the structure sheaf with the left module  $\widehat{R[u]}$ , where  $D$  acts by the scalar derivation determined by  $D(u) = e$ . At every finite level its reduction is the polynomial module  $(R/p^n)[u]$ . The operator  $D = e d/du$  lowers polynomial degree, so it is locally nilpotent on every element of every finite-level reduction. Thus the compatible system defining  $\widehat{R[u]}$  lies in the torsion categories of Proposition 2.14. Notice that this is a levelwise assertion; it does not incorrectly assert literal local nilpotence on an arbitrary restricted series before reduction.

We now justify the use of PBW after completion. For every  $n$ , the ordinary PBW decomposition

$$\mathscr{W}_{R,n} = \bigoplus_{m \geq 0} (R/p^n)[u] D^m$$

shows that right multiplication by  $D$  is injective. All modules in this lemma are left modules. Nevertheless  $r_D : x \mapsto xD$  is a homomorphism of left  $\mathscr{W}_{R,n}$ -modules, since  $r_D(ax) = a r_D(x)$ . Its image  $\mathscr{W}_{R,n} D$  is a left ideal, and PBW identifies the quotient with  $(R/p^n)[u]$ . In that quotient the left action of  $D$  is the scalar derivation  $e d/du$ , because  $Df(u) = f(u)D + e f'(u)$ . We therefore obtain the following homogeneous free resolution in the category of left modules:

$$0 \longrightarrow \mathscr{W}_{R,n} \xrightarrow{(-)D} \mathscr{W}_{R,n} \longrightarrow (R/p^n)[u] \longrightarrow 0,$$

with the first copy shifted by the degree of  $D$ . All transition maps in this inverse system are surjective. The Mittag-Leffler criterion therefore shows that its inverse limit is still exact; in particular, no  $\varprojlim^1$ -term appears. We obtain an exact triangle of derived  $p$ -complete graded modules

$$\widehat{\mathscr{W}}_R \xrightarrow{(-)D} \widehat{\mathscr{W}}_R \longrightarrow \widehat{R[u]},$$

again with the source shifted.

By (2.23), derived global sections are  $\mathrm{RHom}_{\widehat{\mathscr{W}}_R}^{\mathrm{cont}, \mathrm{gr}}(\widehat{R[u]}, -)$ , since global sections are maps from the unit object. Here the notation means the inverse limit of the ambient finite-level derived Hom complexes. In particular, for every derived  $p$ -complete graded module  $M$ ,

$$\mathrm{RHom}_{\widehat{\mathscr{W}}_R}^{\mathrm{cont}, \mathrm{gr}}(\widehat{\mathscr{W}}_R, M) \simeq M,$$

so applying continuous graded  $\mathrm{RHom}$  to the completed resolution above is legitimate. Under  $\mathrm{Hom}_{\widehat{\mathscr{W}}_R}(\widehat{\mathscr{W}}_R, M) \simeq M$ , a homomorphism  $f$  corresponds to  $m = f(1)$ . Precomposition with  $r_D$  sends it to the element  $f(D) = Df(1) = Dm$ . Thus the differential induced by this left free resolution is  $+D$ , not  $-D$  and not a right-module action. With the character convention in the statement, tensoring by  $\mathcal{O}_{\mathcal{X}}(-r)$  and then taking  $\mathbf{G}_m$ -invariants selects degree  $-r$  of the unshifted graded Hom complex. Applying graded Hom to the completed resolution therefore gives

$$[\widehat{R[u]}]_{-r} \xrightarrow{D} [\widehat{R[u]}]_{-r+1} = [Ru^r \xrightarrow{D} Ru^{r-1}].$$

The last equality uses that a fixed homogeneous component is already  $p$ -complete:  $\widehat{R[u]}_{-r} = Ru^r$ . The Leibniz rule and  $D(1) = 0$  give  $D(u^r) = r e u^{r-1}$ , completing the calculation.  $\square$

### 2.5.2. Functoriality of the central Rees comparison.

**Lemma 2.17** (Associated graded commutes with filtered completion). *Let  $\mathcal{D}$  be a stable infinity-category admitting inverse limits, and let  $F_\bullet : \mathbf{N}^{\text{op}} \rightarrow \mathcal{D}$  be a decreasing filtered object. Write*

$$\widehat{F}_r = \varprojlim_{m > r} \text{cofib}(F_m \longrightarrow F_r)$$

for its filtered completion. There is a natural equivalence

$$\text{cofib}(F_{r+1} \longrightarrow F_r) \xrightarrow{\sim} \text{cofib}(\widehat{F}_{r+1} \longrightarrow \widehat{F}_r).$$

The equivalence is natural in morphisms of filtered objects and therefore also in their arrow category.

*Proof.* For  $m > r + 1$ , set

$$Q_{m,r} = \text{cofib}(F_m \longrightarrow F_r).$$

The octahedral diagram for  $F_m \rightarrow F_{r+1} \rightarrow F_r$  gives a natural cofiber sequence

$$Q_{m,r+1} \longrightarrow Q_{m,r} \longrightarrow \text{cofib}(F_{r+1} \longrightarrow F_r).$$

The third term is constant as  $m$  varies. In a stable infinity-category, inverse limits are exact: a finite cofiber is also a finite limit, and limits commute with limits. Taking the inverse limit in  $m$  therefore gives

$$\widehat{F}_{r+1} \longrightarrow \widehat{F}_r \longrightarrow \text{cofib}(F_{r+1} \longrightarrow F_r).$$

This is the asserted equivalence. Every arrow used in the construction is functorial in  $F_\bullet$ ; applying  $\text{Fun}(\Delta^1, -)$  proves the final assertion.  $\square$

**Definition 2.18** (Marked package arrows and their realizations). Fix the Tate index  $i$ .

(1) Package arrows. For  $S = \text{WCart}$  or  $S = R^\Delta$ , let  $\mathcal{A}_i(S)$  be the stable infinity-category of base-change-compatible complete filtered AKN package squares of twist  $i$  on  $S$ . On an oriented transversal prism  $(B, I_B)$ , such an object is a diagram in

$$\text{Fun}(\mathbf{N}^{\text{op}} \times (\Delta^1 \times \Delta^1), \mathcal{D}(B));$$

the transition maps in the filtration and the base-change identifications under morphisms of prisms are part of the object. Let  $\mathcal{A}_i(R^\Delta)^G$  denote the category of  $G$ -equivariant objects.

Define the category of marked package arrows by

$$(2.25) \quad \mathcal{P}_i := \mathcal{A}_i(\text{WCart}) \times_{\mathcal{A}_i(R^\Delta)^G} \text{Fun}(\Delta^1, \mathcal{A}_i(R^\Delta)^G).$$

The functor from the first factor is  $\pi_R^*$ , and the functor from the second factor is evaluation at the source of  $\Delta^1$ . Thus an object of  $\mathcal{P}_i$  is a triple

$$\mathfrak{a} = (\mathcal{F}, a, \mathcal{J}), \quad a : \pi_R^* \mathcal{F} \longrightarrow \mathcal{J},$$

where  $a$  is a  $G$ -equivariant morphism of complete filtered package squares. In particular,  $\mathfrak{a}$  retains the commutativity 2-simplices, the filtration transition maps, and all crystal base-change identifications. The package morphism induced by  $m_x : T_x \rightarrow \overline{B}$  in (2.19) defines such an object.

(2) Filtered squares of marked arrows. Put

$$\mathcal{Q}_i := \text{Fun}(\mathbf{N}^{\text{op}} \times (\Delta^1 \times \Delta^1), \text{Fun}(\Delta^1, \mathcal{D}(\mathbf{Z}_p))).$$

An object of  $\mathcal{Q}_i$  is therefore a filtered adjacent square in which every vertex is itself a marked arrow. Let

$$\text{Fib}_{\text{ad}} : \mathcal{Q}_i \longrightarrow \text{Fun}(\mathbf{N}^{\text{op}}, \text{Fun}(\Delta^1, \mathcal{D}(\mathbf{Z}_p)))$$

be the exact functor which takes the horizontal fiber of the adjacent square at each filtration level. The commutativity 2-simplex of the square supplies the induced arrow between the two fibers. We use the same notation for this exact construction before applying global sections. For the package used below, Proposition 2.9 identifies  $\text{Fib}_{\text{ad}}$  with the octahedral fiber defining the Nygaard object.

(3) Liu realization. For  $\mathcal{M} \in \mathcal{D}(R^\Delta)$ , set

$$\mathcal{K}_X(\mathcal{M}) := \text{fib} \left( \rho^* \mathcal{M} \xrightarrow{\partial_{\mathcal{M}}^{\text{ar}}} \rho^* \mathcal{M} \otimes_A A\epsilon_{\text{ar}} \right).$$

For  $\mathfrak{a} = (\mathcal{F}, a, \mathcal{J})$ , define the marked arrow

$$(2.26) \quad \mathcal{K}_{\text{ev}}(\mathfrak{a}) : \mathcal{K}_X(\pi_R^* \mathcal{F})^{hG} \xrightarrow{\mathcal{K}_X(a)^{hG}} \mathcal{K}_X(\mathcal{J})^{hG} \xrightarrow{\text{res}} \mathcal{K}_X(\mathcal{J}).$$

The symbol  $\text{res}$  denotes the forgetful map from homotopy fixed points to the underlying object; it is not an averaging or trace map. Since homotopy fixed points are exact,

$$\mathcal{K}_X(\pi_R^* \mathcal{F})^{hG} \simeq \text{fib} \left( (\rho^* \pi_R^* \mathcal{F})^{hG} \longrightarrow (\rho^* \pi_R^* \mathcal{F} \otimes_A A\epsilon_{\text{ar}})^{hG} \right).$$

Hence the source of (2.26) is the complex appearing in Liu's coefficient formula.

The parallel global-sections arrow is

$$(2.27) \quad \Gamma_{\text{ev}}(\mathfrak{a}) : \text{R}\Gamma(\text{WCart}, \mathcal{F}) \longrightarrow \text{R}\Gamma([R^\Delta/G], \mathcal{J}) \xrightarrow{\text{res}} \text{R}\Gamma(R^\Delta, \mathcal{J}).$$

The first map is the equivalence

$$\text{RHom}_{\text{WCart}}(\mathcal{O}, \mathcal{F}) \xrightarrow{\sim} \text{RHom}_{[R^\Delta/G]}(\pi_R^* \mathcal{O}, \pi_R^* \mathcal{F})$$

obtained from the full faithfulness in [3, Theorem 3.5.2]; after identifying both terms with global sections, it is followed by the map induced by  $a$ . Applying (2.26) and (2.27) at every vertex defines exact functors

$$\mathcal{K}_{\text{ev}}, \Gamma_{\text{ev}} : \mathcal{P}_i \longrightarrow \mathcal{Q}_i.$$

(4) The two final realizations. Let  $\text{Comp}$  denote the filtered-completion functor. Define

$$\mathbf{L} := \text{Comp} \circ \text{Fib}_{\text{ad}} \circ \mathcal{K}_{\text{ev}}.$$

Let  $\mathcal{Z}_{\text{cen}} = (R^{\text{Nyg}})_{t=0}$ . For  $r \geq 0$ , define the central Rees realization

$$\mathbf{R}_r := \text{R}\Gamma(\mathcal{Z}_{\text{cen}}, -) \circ \text{wt}_{-r} \circ i_0^* \circ \text{Rees} \circ \text{Fib}_{\text{ad}}.$$

Here Rees formation and restriction to  $t = 0$  are applied to the base-change-compatible package before descent.

**Lemma 2.19** (Naturality of Liu's coefficient realization). *There is a natural equivalence*

$$\eta_{\text{ev}} : \Gamma_{\text{ev}} \xrightarrow{\sim} \mathcal{K}_{\text{ev}}$$

in  $\mathcal{Q}_i$ . After applying  $\text{Fib}_{\text{ad}}$ , this equivalence carries the commutativity 2-simplex of a marked package arrow to the specified nullhomotopy used in Proposition 2.9.

*Proof.* Let  $\mathfrak{a} = (\mathcal{F}, a, \mathcal{J}) \in \mathcal{P}_i$ . Liu's arithmetic operator is functorial in the coefficient by [3, Proposition 3.1.21]. The canonical morphism from global sections to its fiber is [3, Proposition 3.1.23], and [3, Proposition 3.3.1] proves that this morphism is an equivalence when  $p > 2$ . For a  $G$ -equivariant coefficient it is  $G$ -equivariant. Taking homotopy fixed points, and then using Liu's finite-descent equivalence [3, Theorem 3.5.2], gives the vertical equivalences in the square

$$(2.28) \quad \begin{array}{ccc} \text{R}\Gamma(\text{WCart}, \mathcal{F}) & \xrightarrow{\Gamma_{\text{ev}}(\mathfrak{a})} & \text{R}\Gamma(R^\Delta, \mathcal{J}) \\ \eta_{\mathcal{F}} \Big\downarrow \sim & & \sim \Big\downarrow \eta_{\mathcal{J}} \\ \mathcal{K}_X(\pi_R^* \mathcal{F})^{hG} & \xrightarrow{\mathcal{K}_{\text{ev}}(\mathfrak{a})} & \mathcal{K}_X(\mathcal{J}). \end{array}$$

The top horizontal arrow includes the restriction map in (2.27); the bottom horizontal arrow includes the restriction map in (2.26). Naturality of Liu's coefficient morphism gives the commutativity 2-simplex before restriction. A limit defining homotopy fixed points is functorial, so the same

2-simplex remains compatible with both restriction maps. This proves that (2.28) is a natural equivalence of marked arrows.

The construction is vertexwise in the filtration and in the adjacent square, hence defines  $\eta_{\text{ev}}$  in  $\mathcal{Q}_i$ . Finally,  $\text{Fib}_{\text{ad}}$  is an exact functor on the enhanced arrow category. It therefore carries the displayed 2-simplex to the induced 2-simplex on horizontal fibers. By Proposition 2.9, that induced 2-simplex is the specified adjacent nullhomotopy.  $\square$

**Lemma 2.20** (Naturality of the AKN–Rees realization). *For every  $r \geq 0$ , there is a natural equivalence of marked arrows*

$$(2.29) \quad \chi_r : \Gamma_{\text{ev}}(\text{gr}^r \text{Fib}_{\text{ad}}(-)) \xrightarrow{\sim} R_r.$$

*In particular, the comparison is natural in the package morphism  $a$ .*

*Proof.* The relative and absolute comparisons in [1, Theorem 1.2(2)–(3)] are natural equivalences of complete filtered packages. Construction 10.44 of [1] makes this naturality explicit for package morphisms. Evaluating those comparisons on

$$a : \pi_R^* \mathcal{F} \longrightarrow \mathcal{J}$$

therefore gives an equivalence in the arrow category.

Bhatt’s Rees equivalence [6, Proposition 2.2.6(3) and Remark 2.2.10] is likewise natural for filtered morphisms. Composing it with the AKN package comparison gives (2.29). To record its arrow-level content, let

$$s, t : \text{Fun}(\Delta^1, \mathcal{D}(\mathbf{Z}_p)) \longrightarrow \mathcal{D}(\mathbf{Z}_p)$$

denote source and target evaluation. At  $\mathbf{a} = (\mathcal{F}, a, \mathcal{J})$ , the naturality 2-cell is

$$(2.30) \quad \begin{array}{ccc} s \Gamma_{\text{ev}}(\text{gr}^r \text{Fib}_{\text{ad}}(\mathbf{a})) & \xrightarrow{\quad} & t \Gamma_{\text{ev}}(\text{gr}^r \text{Fib}_{\text{ad}}(\mathbf{a})) \\ \chi_{r,\mathbf{a}}^{\text{src}} \downarrow \sim & & \sim \downarrow \chi_{r,\mathbf{a}}^{\text{tgt}} \\ s R_r(\mathbf{a}) & \xrightarrow{\text{R}\Gamma(\text{wt}_{-r} i_0^* \text{Rees}(\text{Fib}_{\text{ad}} a))} & t R_r(\mathbf{a}). \end{array}$$

The left vertical map is the absolute AKN comparison followed by the Rees associated-graded equivalence. The right vertical map is the relative base-prism comparison followed by the same Rees equivalence. The commutativity 2-simplex is the value on  $a$  of AKN’s natural transformation. Thus both horizontal maps in (2.30) are realizations of the same package morphism. This proves the asserted natural equivalence of marked arrows.  $\square$

**Proposition 2.21** (Central Rees comparison). *For every  $r \geq 0$ , there is a natural equivalence of arrow-valued functors*

$$(2.31) \quad \Xi_r : \text{gr}^r \mathbf{L} \xrightarrow{\sim} R_r \quad \text{in} \quad \text{Fun}(\mathcal{P}_i, \text{Fun}(\Delta^1, \mathcal{D}(\mathbf{Z}_p))).$$

*It is compatible with the nullhomotopy in the adjacent Nygaard sequence and with base change on the full Čech nerve of the Rees chart. On Pham’s central atlas the corresponding Rees coordinate is  $u = dt^{-1}$ .*

*Proof.* By Definition 2.18,

$$\text{gr}^r \mathbf{L} = \text{gr}^r \text{Comp}(\text{Fib}_{\text{ad}} \mathcal{K}_{\text{ev}}(-)).$$

The required comparison is the following composite of natural transformations in the enhanced arrow category:

$$(2.32) \quad \mathrm{gr}^r \mathrm{Comp}(\mathrm{Fib}_{\mathrm{ad}} \mathcal{K}_{\mathrm{ev}}(-)) \xrightarrow[\eta_{\mathrm{ev}}^{-1}]{\sim} \mathrm{gr}^r \mathrm{Comp}(\mathrm{Fib}_{\mathrm{ad}} \Gamma_{\mathrm{ev}}(-))$$

$$(2.33) \quad \xrightarrow{\sim} \mathrm{gr}^r \mathrm{Fib}_{\mathrm{ad}} \Gamma_{\mathrm{ev}}(-)$$

$$(2.34) \quad \xrightarrow{\sim} \Gamma_{\mathrm{ev}}(\mathrm{gr}^r \mathrm{Fib}_{\mathrm{ad}}(-)) \xrightarrow{\chi_r} \mathbf{R}_r.$$

The first equivalence is Lemma 2.19. The second is Lemma 2.17, applied in the arrow category. For the third, global sections, restriction, and  $\mathrm{Fib}_{\mathrm{ad}}$  are exact; hence they commute with the finite cofiber defining  $\mathrm{gr}^r$ . The last equivalence is Lemma 2.20. This constructs a specified natural equivalence

$$(2.35) \quad \Xi_r : \mathrm{gr}^r \mathbf{L} \xrightarrow{\sim} \mathbf{R}_r.$$

Compatibility with Liu's coefficient equivalence and with finite  $G$ -invariants is built into (2.32). Homotopy fixed points are exact because they are limits in a stable infinity-category. Since  $|G| = p-1$  is invertible in  $\mathbf{Z}_p$ , they agree with the ordinary invariants used elsewhere in the paper. The map denoted  $\mathrm{res}$  remains the forgetful map throughout; no trace factor is introduced.

Compatibility with the adjacent nullhomotopy follows from the final assertion of Lemma 2.19. Compatibility with completion and associated graded is precisely (2.33). These comparisons are therefore defined before any atlas is chosen.

All package transformations in (2.32)–(2.34) commute with base change in  $(B, I_B, x)$ . More precisely, the functoriality of the Nygaard-filtered prismatic package and of its two syntomic structure maps is [1, Theorem 1.2 and Construction 10.44]; the required naturality of Liu's one-correspondence operator is [3, Proposition 3.1.21]; and the functoriality of the Rees construction for filtered maps is [6, Proposition 2.2.6(3) and Remark 2.2.10]. The boundedness and completeness hypotheses of these results hold because every level is obtained by base change from the bounded oriented prism  $(A, (d))$ , and all filtered objects in  $\mathcal{P}_i$  are completed by definition. Applying the construction at every level of the full Rees Čech nerve gives an equivalence of cosimplicial arrow objects. Totalization is functorial, so  $\Xi_r$  descends and retains its commutativity 2-simplices.

Finally pull the descended arrow back to Pham's affine central atlas. On an oriented Rees chart, the affine coordinate is the value of  $I \rightarrow L$  on the chosen generator  $d$ ; hence it is

$$u = dt^{-1}.$$

Pullback is a functor and therefore preserves  $\Xi_r$ . Evaluate  $\Xi_r$  on the marked package arrow induced by  $m_x : T_x \rightarrow \overline{B}$ . Proposition 2.9 identifies its Liu realization with the arrow  $\mathrm{gr}^r(\theta_{\bullet, i})$ , while the definition of  $\mathbf{R}_r$  identifies its other realization with the direct central-Rees coefficient arrow. This proves the assertion.  $\square$

**Lemma 2.22** (Evaluation on the oriented generator). *Under the central-stack identification of Lemma 2.16 and the bases used in (2.16)–(2.17), the degree-zero component of  $\mathrm{gr}^r(\theta_{\bullet, i})$  is*

$$Ru^r \longrightarrow d^r A / d^{r+1} A, \quad u^r \longmapsto [d^r].$$

Here  $u$  is Pham's central-atlas coordinate, identified with the Rees coordinate  $u_q = dt^{-1}$  by the fixed orientation  $I = (d)$ .

*Proof.* Let

$$\mathcal{R}_d = [\mathrm{Spf} A\langle u_q, t \rangle / (u_q t - d) / \mathbf{G}_m] \longrightarrow R^{\mathrm{Nyg}}$$

be Bhatt's oriented Rees chart, where  $u_q = dt^{-1}$  has degree  $-1$ . The hypotheses of [6, Remark 5.5.19] hold because  $A$  is  $p$ -torsion free and  $A/p = \mathbf{F}_p[[q-1]]$  is regular. Thus this map is faithfully flat. On its central fiber the weight  $-r$  line is generated by  $u_q^r$ .

We first fix the coordinate comparison, and then identify the arrow. In the construction preceding [5, Proposition 3.4], the restriction of this very Rees chart to  $t = 0$  is identified, using the chosen generator  $d$  of  $I$ , with

$$[\mathbf{A}_-^1/\mathbf{G}_m]_R \longrightarrow (R^{\text{Nyg}})_{t=0}.$$

In general an  $S$ -point consists of a line bundle  $L$  and a factorization  $I \otimes_A S \xrightarrow{u_S} L \xrightarrow{t_S} S$ . After the fpqc base change which trivializes  $L$ , this point lies on the affine atlas  $\mathbf{A}_-^1 \rightarrow [\mathbf{A}_-^1/\mathbf{G}_m]$ , and  $u_S$  is represented by a map  $I \otimes_A S \rightarrow S$ . Pham's affine coordinate  $u$  is then the function  $u_S(d)$ . This is exactly the Rees coordinate  $u_q = dt^{-1}$ ; it is not a second atlas coordinate differing from it by an unspecified scalar. Proposition 3.4 of [5] factors this same map as

$$[\mathbf{A}_-^1/\mathbf{G}_m]_R \longrightarrow [\mathbf{A}_-^1/(\mathbf{G}_a^\sharp \rtimes \mathbf{G}_m)]_R \xrightarrow{\sim} (R^{\text{Nyg}})_{t=0}.$$

Proposition 2.21 identifies  $\text{gr}^r(\theta_{\bullet,i})$ , in the arrow category, with the global-sections realization of the direct central-Rees coefficient map induced by  $m_x : T_x \rightarrow \overline{B}$ . It is therefore enough to calculate that coefficient map on the oriented Rees chart. AKN's fixed-pair comparison identifies the evaluated component of the package morphism with (2.20). The map from finite homotopy fixed points to the chart is the forgetful map and therefore preserves the oriented section.

There is also a direct local algebra check. For an oriented prism  $(B, I_B = (d_B))$ , relative prismatic cohomology of the base pair  $(B, \overline{B})$  is  $B$ , with  $r$ -th Nygaard lattice  $I_B^r$ . The map  $m_x : T_x \rightarrow \overline{B}$  is a morphism over  $B$ , so the induced map to this base relative prism is the identity on  $B$ ; in particular it sends the oriented generator  $d_B^r$  to  $d_B^r$ . Passing to the Rees module divides this generator by  $t^r$  and therefore sends it to  $(d_B t^{-1})^r = u_q^r$ . This calculation takes place before global sections.

This gives the following diagram of degree-zero cochain terms after taking the  $r$ -th associated graded:

$$(*) \quad \begin{array}{ccc} Ru^r & \xrightarrow{\text{gr}^r(\theta_{\bullet,i})^0} & d^r A / d^{r+1} A \\ \downarrow u=u_q \sim & & \downarrow \sim \chi_r \\ Ru_q^r & \xlongequal{\quad} & Ru_q^r, \end{array} \quad \chi_r([ad^r]) = \bar{a} u_q^r.$$

To verify commutativity, work fpqc-locally on an arbitrary central Rees  $S$ -point, trivialize its line bundle  $L$ , and write the resulting affine-atlas coordinate as  $u_S : I \otimes_A S \rightarrow S$ . The calculation below is  $\mathbf{G}_m$ -equivariant, so equality after this faithfully flat base change descends to the original point. The upper composite is induced by  $m_x$  and sends the  $r$ -fold oriented Nygaard generator to  $u_S(d)^r$ . The lower composite evaluates  $u_q^r$  at the same point and also gives  $u_S(d)^r$ . Thus the two natural transformations agree on every test ring  $S$ , including their map on the chosen generator. The Tate generator is the same on both sides, and Lemma 2.15 shows that AKN's shearing contributes no additional character. Diagram  $(*)$  therefore identifies the degree-zero component and gives  $u^r \mapsto [d^r]$ . For  $r = 0$  this is the unital structure map.  $\square$

**Lemma 2.23** (Linearity on the central fiber). *For every  $r \geq 0$ , the morphism  $\text{gr}^r(\theta_{\bullet,i})$  is induced, before taking global sections, by a morphism of quasi-coherent complexes on the central fiber*

$$\mathcal{Z}_{\text{cen}} := (R^{\text{Nyg}})_{t=0}.$$

After pullback to Pham's affine central atlas  $\mathbf{A}_{-,R}^1 \rightarrow \mathcal{Z}_{\text{cen}}$ , the resulting morphism is  $\widehat{R[u]}$ -linear. Consequently, in the functorial two-term models of Proposition 2.14 and (2.16)–(2.17), it is represented by an  $R$ -linear chain map. This assertion applies to both cochain degrees, not only to the degree-zero component of Lemma 2.22.

*Proof.* We keep the construction before the finite-invariant presentation. For an oriented transversal prism  $(B, I_B)$  and a point  $x : R \rightarrow \overline{B}$ , the morphism of bounded  $\delta$ -pairs

$$(B, T_x) \longrightarrow (B, \overline{B})$$



induces the package morphism (2.19). Every component of this morphism is a morphism of quasi-coherent  $B$ -complexes. In particular, the two vertical arrows in the pointwise adjacent square are  $B$ -linear. Their compatibility with multiplication by  $I_B$ , proved in Lemma 2.12, means that these squares form a morphism of filtered quasi-coherent arrow objects.

Apply the Rees construction objectwise to this filtered square, using the base-change description of associated graded in [6, Proposition 2.2.6(3) and Remark 2.2.10]. This does not involve global sections: on an oriented chart it sends a filtered  $B$ -linear map to a graded  $B\langle u_q, t \rangle / (u_q t - d_B)$ -linear map. Restriction to  $t = 0$  is base change along

$$B\langle u_q, t \rangle / (u_q t - d_B) \longrightarrow (B/d_B)[u_q].$$

It therefore gives a morphism of quasi-coherent complexes which is linear over the  $p$ -completion of  $(B/d_B)[u_q]$ . The crystal compatibilities in (2.20) identify these local morphisms under every map of oriented prisms. Hence they agree on the full Čech nerve of the faithfully flat Rees chart of [6, Remark 5.5.19]; fpqc descent glues them to a morphism on  $\mathcal{Z}_{\text{cen}}$ . If  $i_0 : \mathcal{Z}_{\text{cen}} \hookrightarrow R^{\text{Nyg}}$  denotes the central immersion, the map before global sections is the map of horizontal arrows in the following square of quasi-coherent complexes:

$$(\dagger_0) \quad \begin{array}{ccc} i_0^* \text{Rees}(\mathcal{I} \otimes \pi_R^* \mathcal{E}_{\bullet-1,i}) & \longrightarrow & i_0^* \text{Rees}(\pi_R^* \mathcal{E}_{\bullet,i}) \\ i_0^* \text{Rees}(\text{ev}^-) \downarrow & & \downarrow i_0^* \text{Rees}(\text{ev}) \\ i_0^* \text{Rees}(\mathcal{I} \otimes \mathcal{J}_{\bullet-1,i}) & \longrightarrow & i_0^* \text{Rees}(\mathcal{J}_{\bullet,i}). \end{array}$$

The horizontal arrows are the adjacent Nygaard maps, and the vertical arrows are the two components of (2.20). For filtration level zero, the convention  $\mathcal{E}_{-1,i} = \mathcal{E}_{0,i}$  is used while the upper-left factor  $\mathcal{I}$  is retained, exactly as in Lemma 2.12. Every arrow in  $(\dagger_0)$  is  $\mathcal{O}_{\mathcal{Z}_{\text{cen}}}$ -linear. Let  $\mathcal{F}_{r,i} \rightarrow \mathcal{G}_{r,i}$  be the map obtained by taking the horizontal fibers in  $(\dagger_0)$  and then its weight  $-r$  component. The functorial octahedral cancellation of Lemma 2.3 takes place in the same quasi-coherent category and preserves this linearity.

Apply Proposition 2.21 to the package morphism (2.19). Evaluating  $\Xi_r$  on this arrow gives the following diagram in the arrow category:

$$(\dagger) \quad \begin{array}{ccc} \text{R}\Gamma(\mathcal{Z}_{\text{cen}}, \mathcal{F}_{r,i}) & \xrightarrow{\text{R}\Gamma(\overline{\text{ev}}_{r,i})} & \text{R}\Gamma(\mathcal{Z}_{\text{cen}}, \mathcal{G}_{r,i}) \\ \sim \downarrow & & \downarrow \sim \\ \text{gr}^r((N_{\bullet,i})^\wedge) & \xrightarrow{\text{gr}^r(\theta_{\bullet,i})} & \text{gr}^r \mathcal{C}_{\bullet,i}. \end{array}$$

Here  $\overline{\text{ev}}_{r,i}$  is the morphism constructed from  $(\dagger_0)$ . The vertical arrows and their commutativity 2-simplex are the value of the natural transformation (2.35).

The top arrow is a morphism in  $\mathcal{D}_{\text{qc}}(\mathcal{Z}_{\text{cen}})$ , and  $\mathcal{Z}_{\text{cen}}$  is a stack over  $\text{Spf } R$ . It is therefore  $R$ -linear. Pulling it back along Pham's affine atlas makes it  $\widehat{R[u]}$ -linear. Finally, the completed Weyl resolution used in Proposition 2.14 is functorial in morphisms of equivariant modules, so applying its two-term Hom model produces a strict  $R$ -linear chain map. The restriction map from finite invariants in Lemma 2.12 is only a later global-sections presentation; no scalar linearity is inferred from that restriction map.  $\square$

### 2.5.3. Associated graded and filtered reconstruction.

**Proposition 2.24** (Associated graded). *The map  $\text{gr}^r(\theta_{\bullet,i})$  is an equivalence for every  $r \geq 0$ .*

*Proof.* The source of  $\text{gr}^r(\theta_{\bullet,i})$  is the associated graded of the Nygaard object, while its target is  $\text{gr}^r \mathcal{C}_{\bullet,i}$ . We first justify the geometric model used to compute the source. AKN's absolute comparison theorem [1, Theorem 1.2(3) and Remark 6.8] identifies their completed filtered object



with Bhatt–Lurie’s absolute Nygaard filtration. Its associated graded is intrinsic: it is the top fiber sequence in AKN Lemma 6.10. Bhatt’s structure gauge is constructed in [6, Definition 5.3.10 and Example 5.3.12]. For the present prism  $(A, (d)) = (\mathbf{Z}_p[[q-1]], ([p]_q))$ , the hypotheses of [6, Remark 5.5.19] apply directly:  $A$  is  $p$ -torsion free and  $A/p = \mathbf{F}_p[[q-1]]$  is regular. That remark constructs a faithfully flat map

$$\mathcal{R}_d = [\mathrm{Spf} A\langle u_q, t \rangle / (u_q t - d) / \mathbf{G}_m] \longrightarrow R^{\mathrm{Nyg}}.$$

We compute after pullback to this chart and its full Čech nerve and then descend. Thus no special-coordinate Breuil–Kisin hypothesis and no assertion that  $R$  is quasiregular semiperfectoid is used. Together with AKN’s absolute comparison, this identifies the descended Rees filtration with the absolute filtered object used here. The sign and the weight in the next step are fixed by the Rees convention, rather than by a choice of notation: [6, Proposition 2.2.6(3)] gives

$$\mathrm{gr}^r F \simeq R\Gamma(B\mathbf{G}_m, i^* \mathrm{Rees}(F)(-r)),$$

and [6, Remark 2.2.10] applies the same formula to a filtered stack. Consequently the  $r$ -th associated graded is the weight  $-r$  part of the restriction of the structure gauge to  $t = 0$ . This is not a replacement of the absolute Nygaard stack by one Rees chart.

Pham’s presentation [5, Proposition 3.4] identifies this central fiber with

$$\left[ \mathbf{A}_-^1 / (\mathbf{G}_a^\# \rtimes \mathbf{G}_m) \right]_R,$$

where the action in Pham’s Proposition 3.4 is

$$(a, \lambda) \cdot u = e a + \lambda^{-1} u.$$

We record the resulting Ore relation directly, without relying on the displayed presentation in the statement of Pham’s Lemma 3.11. In divided power notation, the  $\mathbf{G}_a^\#$ -coaction associated with  $D$  is

$$m \longmapsto \sum_{n \geq 0} D^n(m) a^{[n]}.$$

Equivariance of multiplication by  $u$ , restricted to  $\lambda = 1$ , and comparison of the coefficient of  $a^{[1]}$  give

$$D(um) = uD(m) + e m.$$

Thus  $Du - uD = e = E'(\pi)$ . The  $\mathbf{G}_m$ -part of the same action gives  $\deg(u) = -1$  and  $\deg(D) = 1$ . This is also the computation in the proof of [5, Lemma 3.11]. Thus Pham’s quotient computation calculates the intrinsic graded object supplied by AKN Lemma 6.10; the full absolute descent remains part of the AKN comparison above.

Lemma 2.16 computes the source at  $r = 0$  as  $R[0]$ . Its target is also  $R[0]$  by (2.16), and the diagonal retraction is the identity on the orientation generator.

For  $r \geq 1$ , Lemma 2.16 computes the source as

$$[Ru^r \xrightarrow{D} Ru^{r-1}] \simeq [R \xrightarrow{r_e} R],$$

while (2.17) computes the target as  $[R \xrightarrow{e[r]_s} R]$ . Lemma 2.22 identifies the comparison map itself: its degree-zero component is the identity in the displayed oriented bases. Lemma 2.23 identifies the associated-graded arrow, before global sections, with a quasi-coherent morphism on the central fiber. In the functorial central two-term model it is therefore a strict  $R$ -linear chain map, independently of the earlier restriction from finite invariants to the full chart. Hence its degree-one component is multiplication by a scalar  $b_r \in R$ . The chain-map identity gives

$$b_r r e = e[r]_s.$$

The ring  $R$  is a domain and  $e \neq 0$ , so  $b_r = [r]_s/r$ . The lifting-the-exponent formula gives  $v_p([r]_s) = v_p(r)$ ; therefore  $[r]_s/r \in \mathbf{Z}_p^\times$ , and hence  $b_r \in R^\times$ . Thus the associated-graded map is

$$\mathrm{gr}^r(\theta_{\bullet,i}) = \left(1, \frac{[r]_s}{r}\right) : [R \xrightarrow{re} R] \longrightarrow [R \xrightarrow{e[r]_s} R].$$

Both components are isomorphisms. This includes the boundary test  $r = p$ : both  $[p]_s$  and  $p$  have valuation 1, so their quotient is a unit.  $\square$

*Remark 2.25* (Weight and sign conventions). The three normalizations used above are related as follows:

$$\text{Nygaard filtration degree } r \quad \longleftrightarrow \quad \text{central } \mathbf{G}_m\text{-weight } -r.$$

In Pham's central atlas the source complex in weight  $-r$  is  $[Ru^r \xrightarrow{+re} Ru^{r-1}]$ , because  $\deg(u) = -1$ ,  $\deg(D) = 1$ , and  $Du - uD = e$ . In the filtered  $q$ -model, after the target trivializations used in (2.17), the corresponding complex is  $[R \xrightarrow{+e[r]_s} R]$ . The degree-zero bases  $u^r$  and  $[d^r]$  are identified by the evaluation map of Lemma 2.22; its degree-zero scalar is (1), and the degree-one comparison is the unit  $[r]_s/r$ . These conventions fix both the weight and the signs used in Proposition 2.24.

**Theorem 2.26** (The filtered  $q$ -model). *For every  $r \geq 0$ , relative to the cyclotomic coordinate, orientation, and Tate-line trivialization fixed in Theorem 1.1, the map of Proposition 2.13 is a canonical equivalence*

$$((N_{\bullet,i})^\wedge)_r \simeq \mathcal{C}_{r,i}.$$

*In particular,*

$$\begin{aligned} \widehat{\mathbb{A}}_R^{(1)}\{i\} &\simeq [A \xrightarrow{\tilde{D}_i} A\omega], \\ \mathrm{N}^{\geq i} \widehat{\mathbb{A}}_R^{(1)}\{i\} &\simeq [d^i A \xrightarrow{\tilde{D}_i} d^{i-1} A\omega]. \end{aligned}$$

*Proof.* The filtered  $q$ -complex is derived complete. To justify the classical completeness used here, let  $\mathfrak{m} = (p, q-1)$ . The ring  $A$  is Noetherian and  $\mathfrak{m}$ -adically complete,  $d \in \mathfrak{m}$ , and each ideal  $d^n A$  is  $\mathfrak{m}$ -adically closed by Krull intersection applied to  $A/d^n A$ . Compatible lifts of an element of  $\varprojlim_n A/d^n$  differ successively by elements of  $d^n A \subseteq \mathfrak{m}^n$ , hence form an  $\mathfrak{m}$ -adic Cauchy sequence. Its limit has the prescribed image modulo every  $d^n$  by the closedness just noted. Thus

$$A \xrightarrow{\sim} \varprojlim_n A/d^n.$$

Since  $A\omega$  is finite free over  $A$ , both terms are derived  $d$ -complete; here we use that, over a Noetherian ring, classical and derived  $d$ -completeness agree for finitely generated modules. Thus the canonical maps to the derived inverse limits of their quotients by  $d^n$  are equivalences. In degree one the filtration is

$$A\omega = A\omega \supset dA\omega \supset d^2 A\omega \supset \cdots.$$

The repeated initial term only says that the degree-one part of  $\mathrm{gr}^0$  vanishes. Removing that repetition leaves the ordinary  $d$ -adic tower, so it introduces neither an additional associated-graded piece nor a derived inverse-limit term. Both filtrations are separated by Krull intersection. The source is complete by the definition of Nygaard completion. The cone of  $\theta_{\bullet,i}$  is therefore complete and has zero associated graded by Proposition 2.24. Its finite filtered quotients vanish by induction on the filtration length, and passage to the inverse limit shows that the cone vanishes.  $\square$

3. THE SYNTOMIC  $q$ -OPERATOR SQUARE

AKN define relative Frobenius, the comparison map to Nygaard completion, and their composite  $c\varphi$  in Definitions 7.1, 7.2, and 7.7 and Lemma 7.5 of [1]. Their syntomic map is  $\text{can} - c\varphi$  [1, Definition 7.8]. Over the base  $\mathbf{Z}_p$ , the absolute comparison [1, Theorem 1.2(3)] identifies this package with the absolute prismatic package of Bhatt–Lurie. Under this identification,  $c\varphi$  is the completed divided Frobenius  $\widehat{\varphi}\{i\}$  of [2, Notation 5.7.5 and Remark 5.7.10]. Multiplication by  $-1$  on the target identifies the fiber of  $\text{can} - c\varphi$  with the fiber of  $c\varphi - \text{can}$ .

## 3.1. The canonical map and divided Frobenius.

**Lemma 3.1** (A semilinear model for the syntomic map). *On the oriented base pair  $(A, R) = (A, A/(d))$ , the syntomic package map  $c_{R/A;0,i} \circ \varphi_{R/A;i,i}$  induces, before taking either fiber, a strict morphism of two-term arrow objects*

$$(\ddagger) \quad \begin{array}{ccc} d^i A & \xrightarrow{\tilde{D}_i|_{d^i A}} & d^{i-1} A\omega \\ \Phi_i^0 \downarrow & & \downarrow \Phi_i^1 \\ A & \xrightarrow{\tilde{D}_i} & A\omega. \end{array}$$

After trivializing the upper terms by  $d^i a \leftrightarrow a$  and  $d^{i-1} b\omega \leftrightarrow b$ , its upper horizontal arrow is  $D_i$ . Both vertical maps are  $\varphi_A$ -semilinear maps of  $p$ -complete  $A$ -modules. The image of  $(\ddagger)$  in the derived arrow category is the map  $\Phi_C$  of Proposition 2.13.

*Proof.* For the base pair, relative prismatic cohomology is the discrete ring  $A$ , and every oriented Breuil–Kisin and Nygaard term occurring here is an invertible rank-one  $A$ -module. Thus the relative divided Frobenius and the map  $c$  are maps of discrete  $p$ -complete modules: the first is linear for its displayed source, while the second is  $\varphi_A$ -semilinear. Their composite has semilinear components  $\Phi_i^0$  and  $\Phi_i^1$ .

Liu’s one-correspondence construction [3, Proposition 3.1.21] attaches the arithmetic operator functorially to a coefficient map. Applied to the two components of this package morphism, that functoriality supplies a morphism of the two operator arrows, including its specified commutative square. In the present base pair all four coefficient objects are discrete modules. If  $M$  and  $N$  are any two such  $\mathbf{Z}_p$ -modules, viewed as objects of  $\mathcal{D}(\mathbf{Z}_p)$ , then for every  $n > 0$  one has

$$\pi_n \text{Map}_{\mathcal{D}(\mathbf{Z}_p)}(M, N) = \text{Hom}_{\mathcal{D}(\mathbf{Z}_p)}(M, N[-n]) = 0.$$

Consequently a homotopy between two maps  $M \rightarrow N$ , if it exists, is unique, and the two maps determine the same homomorphism of ordinary modules. We may therefore represent the specified commutative square by the strictly commuting square of module maps  $(\ddagger)$ . Applying the ordinary two-term fiber construction to that strict representative gives the same arrow object in  $\mathcal{D}(\mathbf{Z}_p)$ . The formulas for its horizontal arrows are the target calculation of Lemma 2.12 together with Theorem 2.26. Finally, Construction 10.44 of AKN identifies this same morphism of arrows with the syntomic-component map used in the package square of Proposition 2.13.  $\square$

**Lemma 3.2** (The cyclotomic target calculation). *On Liu’s universal cyclotomic chart, corresponding to the oriented prismatic pair  $(A, R)$  with  $R = A/(d)$ , the target AKN syntomic component*

$$c_{R/A;0,i} \circ \varphi_{R/A;i,i}$$

has, for  $a, b \in A$ , the formulas

$$d^i a \longmapsto \varphi(a), \quad d^{i-1} b\omega \longmapsto q^{p-1} \varphi(b)\omega.$$

*Proof. Degree zero.* We first identify the filtered map in degree zero. The pair  $(A, R) = (A, A/(d))$  is a prismatic  $\delta$ -pair. The relative comparison [1, Theorem 1.2(2)] identifies the Nygaard-filtered terms of twist  $i$ , together with the maps  $c$  and  $\varphi$ , with their usual relative prismatic models. For a prismatic pair, the proof of AKN Lemma 7.5 constructs  $c$  as a filtered refinement of the unfiltered map  $c_0$ ; this is the case used here. For the base prism  $R = A/(d)$ , relative prismatic cohomology is  $A$ . If  $e_i$  is the oriented Breuil–Kisin generator, then  $\varphi(e_i) = d^{-i}e_i$  by [2, Remark 2.5.9 and Notation 2.6.3]. Consequently the relative divided Frobenius of AKN Definition 7.1 is  $A$ -linear and sends  $d^i a e_i^{(1)}$  to  $a e_i$ . In the same compared package, the map  $c$  is  $\varphi_A$ -semilinear and sends the oriented unit generator  $e_i$  to  $e_i^{(1)}$ . The construction in the proof of AKN Lemma 7.5 supplies its filtered refinement in this prismatic case. Hence the degree-zero component of the composite  $c_{R/A;0,i} \circ \varphi_{R/A;i,i}$  is

$$d^i a e_i^{(1)} \xrightarrow{\varphi_{R/A;i,i}} a e_i \xrightarrow{c_{R/A;0,i}} \varphi_A(a) e_i^{(1)}.$$

Here  $e_i^{(1)}$  denotes the generator on the Frobenius-twisted source and target, while  $e_i$  denotes the generator of the untwisted intermediate prismatic term. The relative divided Frobenius in the first arrow is  $A$ -linear; the second arrow is the  $\varphi_A$ -semilinear map which contributes the coefficient  $\varphi_A(a)$ . The factor  $d^i$  is the Nygaard lattice which clears the pole  $d^{-i}$  of the twist Frobenius; the displayed first arrow is the divided Frobenius on that lattice. Thus no residual factor  $\varphi_A(d)^i/d^i$  occurs. The Day-weight-zero target is generated by  $e_i^{(1)}$ . After suppressing the source and target bases  $e_i^{(1)}$ , the displayed composite is precisely  $d^i a \mapsto \varphi_A(a)$ . This records both the Day-convolution index 0 and the Tate index  $i$ , without asserting that forgetting the filtration identifies  $c$  with  $c_0$  for an arbitrary bounded  $\delta$ -pair.

*Degree one.* We determine the degree-one coefficient in the strict arrow model of Lemma 3.1. The relative Frobenius in AKN Definition 7.1 is linear for its displayed source module, the map  $c$  of AKN Lemma 7.5 is  $\varphi_A$ -semilinear, and every evaluation and comparison map used to transport the package is  $A$ -linear. Hence the transported composite is  $\varphi_A$ -semilinear in both cochain degrees; Lemma 3.1 shows that these are the components of a chain map. By Theorem 2.26, its source and target in degree one are free rank-one  $A$ -modules. Since the composite is  $\varphi$ -semilinear, there is a unique  $\mu_i \in A$  such that its degree-one component is

$$d^{i-1} b \omega \mapsto \mu_i \varphi(b) \omega.$$

The chain-map identity gives, for every  $a \in A$ ,

$$(3.1) \quad \tilde{D}_i(\varphi(a)) = \mu_i \varphi(D_i(a)).$$

On the other hand,  $\tilde{r}_i \varphi = \varphi r_i$  and

$$\varphi(h) = q^p(q^p - 1) = q^{p-1} d h, \quad h = q(q - 1),$$

so a direct calculation gives

$$(3.2) \quad \begin{aligned} \tilde{D}_i(\varphi(a)) &= \frac{\varphi(r_i(a) - a)}{d h} = \frac{\varphi(h)}{d h} \varphi(D_i(a)) \\ &= q^{p-1} \varphi(D_i(a)). \end{aligned}$$

It remains only to justify cancellation. Since

$$\lambda(q) = 1 - \frac{p}{2}(q - 1) + O((q - 1)^2),$$

the series  $\lambda(q)^i - 1$  is nonzero for  $i \geq 1$ . Thus  $D_i(1) = (\lambda(q)^i - 1)/h \neq 0$ . The ring  $A$  is a domain and  $\varphi : A \rightarrow A$  is injective. Comparing (3.1) with (3.2) at  $a = 1$  therefore yields  $\mu_i = q^{p-1}$ , as required.

For comparison, GLSQ [4, Remark 10.2(2)–(3), especially equation (51)] obtains the same coefficient from the differential identity  $\varphi(\omega) = d q^{p-1} \omega$ .  $\square$

**Proposition 3.3** (Structural maps in the completed  $q$ -model). *Under Theorem 2.26, the canonical map is the inclusion of the two lattices. The divided Frobenius is*

$$\begin{aligned} d^i a &\longmapsto \varphi(a), \\ d^{i-1} b \omega &\longmapsto q^{p-1} \varphi(b) \omega. \end{aligned}$$

After trivializing the source by  $d^i, d^{i-1} \omega$  and the target by  $1, \omega$ , the difference  $c\varphi - \text{can}$  is  $F_i = \varphi - d^i$  in degree zero and  $V_i = q^{p-1} \varphi - d^{i-1}$  in degree one.

*Proof.* The assertion for  $\text{can}$  follows because  $\theta_{\bullet, i}$  is a morphism of filtered objects. For divided Frobenius, the pointwise comparison is the morphism of packages (2.19). We use only its syntomic-component square, where the relevant horizontal map is  $c_{S/B;0,i} \circ \varphi_{S/B;i,i}$ ; commutativity follows directly from the functoriality of  $c\varphi$  in AKN Construction 10.44. After restriction to Liu's universal cyclotomic chart, Lemma 3.2 computes the target member of this package square. Functoriality of (2.19) is used only to transport that target calculation to the source member. The exact operations used in Proposition 2.13 and the subsequent completion transport that square to  $\theta_{\bullet, i}$ , so the same two formulas hold in the completed model.

Subtracting the lattice inclusions gives the two claimed formulas. Their horizontality is the algebraic identity

$$\tilde{D}_i(\varphi(a)) = q^{p-1} \varphi(D_i(a)), \quad \tilde{D}_i(d^i a) = d^{i-1} D_i(a).$$

□

**Corollary 3.4** (The syntomic  $q$ -operator square). *There is a natural equivalence*

$$\mathbf{Z}_p(i)(R) \simeq \text{TotFib} \begin{array}{ccc} A & \xrightarrow{D_i} & A \\ \varphi - d^i \downarrow & & \downarrow q^{p-1} \varphi - d^{i-1} \\ A & \xrightarrow{\tilde{D}_i} & A. \end{array}$$

With the convention  $c\varphi - \text{can}$  for the vertical maps, this is the complex  $C_i^\bullet$  of (1.3).

*Proof.* By AKN Definition 7.8 and the absolute comparison of [1, Theorem 1.2(3)],  $\mathbf{Z}_p(i)(R)$  is the fiber of  $\text{can} - c\varphi$  from filtration degree  $i$  to degree zero. Multiplication by  $-1$  on the target replaces this map by  $c\varphi - \text{can}$ . Use the degree- $i$  and degree-zero cases of Theorem 2.26 and Proposition 3.3. Explicitly, if  $f : [X^0 \rightarrow X^1] \rightarrow [Y^0 \rightarrow Y^1]$  is a chain map, then  $\text{Cone}(f)[-1]$  is

$$[X^0 \xrightarrow{(d_X, f^0)} X^1 \oplus Y^0 \xrightarrow{(f^1, -d_Y)} Y^1].$$

Applying this formula gives (1.3). The square commutes by the two identities in Lemma 2.11; hence the displayed second differential annihilates the first. □

#### 4. DIRECT COMPUTATION OF THE COHOMOLOGY

Let  $C_i^\bullet$  denote the three-term complex in (1.3). Put

$$T = q - 1, \quad s = p + 1, \quad W = q^p - 1 = dT.$$

We compute its cohomology by a finite Herr reduction. No syntomic-étale comparison enters the calculation. Every argument in this section is algebraic in  $A$  and in the four operators of (1.1).

**4.1. Finite Herr reduction.** For  $m \geq 0$ , inside the common fraction field of  $A$ , set

$$\mathcal{M}_m = T^{-m}A, \quad \mathcal{N}_m = W^{-m}A, \quad K_m^\bullet = \left[ \mathcal{M}_m \xrightarrow{\varphi-1} \mathcal{N}_m \right].$$

The complex is placed in degrees 0, 1. Since  $\varphi$  and  $\gamma$  commute, the operator

$$P_i := s^i \gamma - 1$$

defines a chain map  $K_i^\bullet \rightarrow K_{i-1}^\bullet$ . We use the cohomological totalization convention

$$\text{Tot}^0 = K_i^0, \quad \text{Tot}^1 = K_{i-1}^0 \oplus K_i^1, \quad \text{Tot}^2 = K_{i-1}^1.$$

Writing  $d_K = \varphi - 1$ , its two differentials are

$$x \mapsto (P_i x, d_K x), \quad (y, z) \mapsto d_K y - P_i z.$$

**Proposition 4.1** (Exact Herr form of the square). *There is an isomorphism of complexes*

$$C_i^\bullet \cong \text{Tot}(K_i^\bullet \xrightarrow{P_i} K_{i-1}^\bullet).$$

In the four corners it is given by

$$a \mapsto T^{-i}a, \quad b \mapsto qT^{1-i}b, \quad c \mapsto W^{-i}c, \quad z \mapsto qW^{1-i}z.$$

*Proof.* Since

$$\gamma(T) = [s]_q T, \quad \gamma(W) = [s]_{q^p} W,$$

the definitions of  $r_i$  and  $\tilde{r}_i$  give

$$\begin{aligned} (s^i \gamma - 1)(T^{-i}a) &= qT^{1-i}D_i(a), \\ (s^i \gamma - 1)(W^{-i}c) &= qW^{1-i}\tilde{D}_i(c). \end{aligned}$$

Moreover,

$$\begin{aligned} (\varphi - 1)(T^{-i}a) &= W^{-i}F_i(a), \\ (\varphi - 1)(qT^{1-i}b) &= qW^{1-i}V_i(b). \end{aligned}$$

These identities identify both total differentials, including their signs. □

**Lemma 4.2** (Degree-zero Herr cohomology). *For every  $m \geq 0$ ,*

$$H^0(K_m^\bullet) = \mathbf{Z}_p.$$

*Under these identifications, the map induced by  $P_i = s^i \gamma - 1$  is multiplication by  $s^i - 1$ .*

*Proof.* An element of  $H^0(K_m^\bullet)$  has the form  $T^{-m}a$  with

$$\varphi(a) = d^m a.$$

If  $a \neq 0$ , write  $a = T^r u$ , where  $u(0) \neq 0$ . Since  $\varphi(T) = W = dT$ , comparison of constant terms after cancelling  $T^r$  gives  $p^r u(0) = p^m u(0)$ , and hence  $r = m$ . The remaining equation is  $\varphi(u) = u$ . If  $u_n T^n$  is the first nonconstant term of  $u$ , the coefficient of  $T^n$  in  $\varphi(u) - u$  is  $(p^n - 1)u_n$ . This is zero only when  $u_n = 0$ , because  $p^n - 1 \in \mathbf{Z}_p^\times$ . Induction on  $n$  gives  $u \in \mathbf{Z}_p$ . Thus  $T^{-m}a$  is constant. Conversely, constants are fixed by  $\varphi$ . Finally,  $\gamma$  fixes constants, so  $P_i$  acts on this copy of  $\mathbf{Z}_p$  by  $s^i - 1$ . □

Multiplication by  $W^m$  identifies  $H^1(K_m^\bullet)$  with

$$Q_m := A/F_m A, \quad F_m = \varphi - d^m.$$

Here  $Q_m$  is the cokernel in the category of  $\mathbf{Z}_p$ -modules;  $F_m$  is  $\mathbf{Z}_p$ -linear but not  $A$ -linear. Under this identification, the map induced by  $P_i$  is

$$\bar{L}_i : Q_i \longrightarrow Q_{i-1}, \quad L_i = q\tilde{D}_i = \frac{\tilde{r}_i - 1}{W}.$$

Indeed,  $L_i F_i = F_{i-1}(qD_i)$ .

**Lemma 4.3** (Finite freeness of the vertical cokernels). *For every  $m \geq 0$ , the module  $Q_m$  is finite free over  $\mathbf{Z}_p$ , of rank*

$$\text{rank}_{\mathbf{Z}_p}(Q_m) = 1 + m(p-1).$$

For  $m \geq 1$ , multiplication by  $W$  gives a short exact sequence

$$(4.1) \quad 0 \longrightarrow Q_{m-1} \xrightarrow{[a] \mapsto [Wa]} Q_m \longrightarrow \frac{A/WA}{\mathbf{Z}_p(1-d^m)} \longrightarrow 0.$$

The last term is finite free of rank  $p-1$ .

*Proof.* The operator  $\varphi - 1$  restricts to an automorphism of  $TA$ : in the basis  $T, T^2, \dots$ , its  $n$ -th diagonal coefficient is  $p^n - 1$ , which is a  $p$ -adic unit, and all remaining terms have larger  $T$ -adic order. Successive solution modulo  $T^N$ , followed by the inverse limit using the  $T$ -adic completeness of  $A$ , constructs its inverse on  $TA$ . Hence  $Q_0 = A/(\varphi - 1)A \cong \mathbf{Z}_p$ .

For  $m \geq 1$ , write  $a = a(0) + Tb$ . The identity

$$F_m(Tb) = WF_{m-1}(b)$$

gives

$$(4.2) \quad F_m A = \mathbf{Z}_p(1-d^m) + WF_{m-1}A.$$

Equation (4.2) shows that multiplication by  $W$  induces the first map in (4.1). To check injectivity, suppose

$$Wa = c(1-d^m) + WF_{m-1}(b), \quad c \in \mathbf{Z}_p.$$

The polynomial  $W = (1+T)^p - 1$  is distinguished of degree  $p$ , so Weierstrass division identifies  $A/WA$  with the finite free  $\mathbf{Z}_p$ -module on  $1, T, \dots, T^{p-1}$ . In this basis the class of  $1-d^m$  has first coordinate  $1-p^m \in \mathbf{Z}_p^\times$ . It is therefore a primitive vector: projection to its unit coordinate splits its  $\mathbf{Z}_p$ -span, and in particular no nonzero element of  $\mathbf{Z}_p$  annihilates it. Reducing the displayed equality modulo  $W$  now gives  $c = 0$ . Cancellation of the non-zero-divisor  $W$  then gives  $a \in F_{m-1}A$ .

The cokernel is the last term in (4.1). The polynomial  $W = q^p - 1$  therefore makes  $A/WA$  finite free of rank  $p$ , and the primitive-vector argument just given shows that the indicated  $\mathbf{Z}_p$ -span is a direct summand. The last term is consequently finite free of rank  $p-1$ . Induction on  $m$  proves the assertion.  $\square$

In particular, each image  $F_m A \subset A$  is closed for the  $p$ -adic topology: its quotient  $Q_m$  is finite free over  $\mathbf{Z}_p$ , hence separated. Thus the algebraic cokernels used below agree with the cokernels in  $p$ -complete  $\mathbf{Z}_p$ -modules; no additional completed-cokernel convention is being imposed.

The inclusions in (4.1) define a filtration

$$Q_0 \subset Q_1 \subset \dots \subset Q_m.$$

In the fractional model,  $\gamma$  acts on the quotient  $Q_m/Q_{m-1}$  as multiplication by  $s^{-m}$ . Indeed, on the new principal-part quotient one has  $W = 0$ , so

$$\gamma(q) = q^{p+1} = q, \quad [s]_{q^p} = s,$$

and the factor  $W^{-m}$  contributes  $s^{-m}$ .

**Corollary 4.4** (The kernel of the finite map). *For every  $i \geq 1$ , the map  $\bar{L}_i : Q_i \rightarrow Q_{i-1}$  is surjective after inverting  $p$ . Its integral kernel is finite free of rank  $p-1$ , and its cokernel is finite.*

*Proof.* On  $Q_0 \otimes \mathbf{Q}_p$ , the operator  $P_i = s^i \gamma - 1$  is multiplication by  $s^i - 1$ . On  $(Q_m/Q_{m-1}) \otimes \mathbf{Q}_p$ , for  $1 \leq m \leq i-1$ , it is multiplication by  $s^{i-m} - 1$ . These scalars are nonzero. Thus the restriction of  $P_i$  from  $Q_{i-1} \otimes \mathbf{Q}_p$  to itself is an automorphism, so  $\bar{L}_i \otimes \mathbf{Q}_p$  is surjective. The rank difference in Lemma 4.3 gives a free kernel of rank  $p-1$ .  $\square$



**Proposition 4.5** (Direct computation in degrees zero and one). *For every  $i \geq 1$ ,*

$$H^0(C_i^\bullet) = 0$$

*and, noncanonically as a  $\mathbf{Z}_p$ -module,*

$$H^1(C_i^\bullet) \cong A/(d) \oplus \mathbf{Z}_p/((p+1)^i - 1).$$

*Moreover,*

$$H^2(C_i^\bullet) = \text{coker}(\bar{L}_i).$$

*Proof.* The two-column spectral sequence of Proposition 4.1 has, in vertical degree zero,

$$\mathbf{Z}_p \xrightarrow{s^i - 1} \mathbf{Z}_p,$$

and, in vertical degree one,

$$Q_i \xrightarrow{\bar{L}_i} Q_{i-1}.$$

There are no later differentials. The first map is injective and has cokernel  $\mathbf{Z}_p/(s^i - 1)$ . Corollary 4.4 identifies the kernel of the second map with a free module of rank  $p - 1$ . The extension in degree one splits because its quotient is free. Finally,  $A/(d)$  is free of rank  $p - 1$ , giving the stated noncanonical identification. Degree two is the displayed cokernel.  $\square$

**Corollary 4.6** (Weight one). *For  $i = 1$ ,*

$$H^0(C_1^\bullet) = 0, \quad H^1(C_1^\bullet) \cong A/(d) \oplus \mathbf{Z}_p/p, \quad H^2(C_1^\bullet) = 0.$$

*Proof.* Here  $Q_0 \cong \mathbf{Z}_p$ , with quotient map given by the constant term. The element  $L_1(T)$  has constant term 1. Thus  $\bar{L}_1 : Q_1 \rightarrow Q_0$  is surjective. Apply Proposition 4.5.  $\square$

**4.2. The remaining elementary divisor.** We now compute the finite cokernel without an infinite Smith reduction. The first input is the exact map on the associated graded of the finite  $W$ -filtration.

For later use, observe that for  $1 \leq k \leq i - 1$ ,

$$\tilde{r}_i(W^k a) = s^k W^k \tilde{r}_{i-k}(a).$$

After subtracting  $W^k a$  and dividing by  $W$ , this becomes

$$(4.3) \quad L_i(W^k a) = W^{k-1}((s^k - 1)a + s^k W L_{i-k}(a)).$$

**Lemma 4.7** (At most one Cartier generator). *For  $i \geq 2$ ,*

$$\dim_{\mathbf{F}_p}(\text{coker}(\bar{L}_i)/p) \leq 1.$$

*Proof.* Since  $Q_i$  and  $Q_{i-1}$  are finite free over  $\mathbf{Z}_p$ , reduction modulo  $p$  commutes with the cokernel. Write

$$k = \mathbf{F}_p, \quad B = k[q]/(q^p - 1) = k[T]/(T^p), \quad \bar{Q}_m = Q_m/pQ_m.$$

The exact sequences (4.1), which remain exact after reduction modulo  $p$ , give an increasing filtration

$$\mathcal{F}_r \bar{Q}_i := W^{i-r} \bar{Q}_r, \quad 0 \leq r \leq i.$$

Here and below the right-hand side denotes the image of the iterated injection in (4.1). In particular, the superscript  $W^{i-r}$  records that iterated injection; it is not multiplication by the class of  $W$ , which is zero in the auxiliary ring  $B = A/(p, W)$ . Its graded pieces are

$$(4.4) \quad \text{gr}_0 \bar{Q}_i \cong k, \quad \text{gr}_r \bar{Q}_i \cong B/k(1 - \bar{d}^r) \quad (r \geq 1),$$

where  $\bar{d} = T^{p-1}$ . In particular,  $\bar{d}^r = 0$  in  $B$  for  $r \geq 2$ . On the bottom piece  $\mathcal{F}_0 \bar{Q}_i \cong \bar{Q}_0$ , the map  $\bar{L}_i$  is induced by  $P_i = s^i \gamma - 1$ . The constant-term identification  $Q_0 \cong \mathbf{Z}_p$  makes  $\gamma$  the identity, so this bottom map is zero modulo  $p$ .

For  $1 \leq r < i$ , apply (4.3) with  $k = i - r$ . Since  $s^k - 1 \in p\mathbf{Z}_p$ , reduction modulo  $p$  gives

$$\bar{L}_i(W^{i-r}a) = W^{i-r}\bar{L}_r(a).$$

For  $r = i$ , the same equality is tautological. Consequently  $\bar{L}_i$  lowers the filtration index by one, and the induced map

$$\mathrm{gr}_r \bar{Q}_i \longrightarrow \mathrm{gr}_{r-1} \bar{Q}_{i-1}$$

is the map induced by  $\bar{L}_r$ . On  $\mathrm{gr}_1 \rightarrow \mathrm{gr}_0$  this map is surjective: the constant term of  $L_1(T)$  is 1.

It remains to compute the maps for  $r \geq 2$ . In characteristic  $p$ ,

$$\gamma(q) = q(1 + W), \quad [p + 1]_{q^p} = 1 + W^{p-1} + W^p.$$

Since  $p$  is odd, the second identity implies

$$\left( \frac{p+1}{[p+1]_{q^p}} \right)^r \equiv 1 \pmod{W^2}.$$

Taylor expansion modulo  $W^2$  therefore gives, for  $a \in A$ ,

$$(4.5) \quad \bar{L}_r(a) \equiv q \frac{da}{dq} \pmod{W}.$$

Thus the map on the graded pieces in (4.4) is induced by the Euler operator

$$E = q \frac{d}{dq} : B \longrightarrow B.$$

In the basis  $1, q, \dots, q^{p-1}$ , one has  $E(q^a) = aq^a$ . Hence

$$\ker(E) = k \cdot 1, \quad \mathrm{im}(E) = \langle q, q^2, \dots, q^{p-1} \rangle_k.$$

For  $r \geq 3$ , both the source and target of the graded map are  $B/k \cdot 1$ , and  $E$  induces an isomorphism. For  $r = 2$ , the map is

$$B/k \cdot 1 \longrightarrow B/k(1 - \bar{d}).$$

Since

$$1 - \bar{d} = - \sum_{a=1}^{p-1} q^a \in \mathrm{im}(E),$$

its image has codimension one in the target. We have therefore shown that every graded map is surjective except the map from degree 2 to degree 1, whose cokernel is one-dimensional.

For clarity, the finite-filtered lifting used here is the following elementary argument. Start with a target class and its highest nonzero graded component. Whenever the corresponding graded map is surjective, choose a graded preimage, lift it to the source filtration, and subtract its image; the highest nonzero filtration degree strictly decreases. Because the filtration is finite, this process terminates. At the unique exceptional map, from degree 2 to degree 1, the unliftable part is the image of the target component in its one-dimensional graded cokernel. After choosing one lift of a representative of that cokernel, all lower components are again removed by the surjective graded maps. Thus the total cokernel is a quotient of that one-dimensional exceptional graded cokernel. Hence

$$\dim_k \mathrm{coker}(\bar{L}_i \bmod p) \leq 1,$$

as required.  $\square$

We next construct the integral functional that detects the exponent. Let  $K = \mathbf{Q}_p(\mu_p)$ . If  $m \geq 1$  and  $y \in W^{-m}A$ , define

$$(4.6) \quad \mathrm{Res}_p(y) := \sum_{\zeta^{p=1}} \mathrm{res}_{q=\zeta} \left( y \frac{dq}{q} \right) \in K.$$

**Lemma 4.8** (Integral cyclotomic residue). *For every  $m \geq 1$ , the map in (4.6) takes values in  $\mathbf{Z}_p$ , is continuous, and is surjective. It kills*

$$(\varphi - 1)T^{-m}A \subset W^{-m}A$$

*and therefore induces a surjection*

$$\text{Res}_m^H : H^1(K_m^\bullet) \longrightarrow \mathbf{Z}_p.$$

*Moreover,*

$$\text{Res}_p(\gamma y) = s^{-1} \text{Res}_p(y).$$

*Proof.* It is enough first to take  $y = q^n/W^m$ , with  $n \geq 0$ . Under  $u = q^p$ , summation over the  $p$  local branches gives

$$(4.7) \quad \text{Res}_p \left( \frac{q^n}{W^m} \right) = \begin{cases} \binom{n/p-1}{m-1}, & p \mid n, \\ 0, & p \nmid n. \end{cases}$$

Indeed, the trace of  $q^n$  is zero unless  $p \mid n$ ; if  $n = pr$ , the factor  $p$  in the trace cancels  $dq/q = p^{-1}du/u$ , and the required residue is the coefficient of  $(u-1)^{m-1}$  in  $u^{r-1}$ . The generalized binomial coefficient in (4.7) lies in  $\mathbf{Z}_p$ , including  $n = 0$ . For  $n = 0$  it equals  $(-1)^{m-1}$ , so the map is surjective. Polynomials in  $q$  are dense in  $A$  for the  $(p, T)$ -adic topology. For fixed  $m$ , each local residue depends continuously on the first  $m$  Taylor coefficients at  $T = \zeta - 1$ . More explicitly, translation defines a continuous map

$$A = \mathbf{Z}_p[[T]] \longrightarrow K[[q - \zeta]], \quad f(T) \longmapsto f((\zeta - 1) + (q - \zeta)),$$

from the  $(p, T)$ -adic topology to the coefficientwise topology: the substitution converges because  $\zeta - 1$  is topologically nilpotent. Taking the finite jet of order  $m - 1$ , multiplying by the fixed Laurent expansion of  $W^{-m}dq/q$ , and extracting the coefficient of  $(q - \zeta)^{-1}$  are continuous operations. Summing over the finite set of  $p$ -th roots proves continuity. Density and (4.7) then prove integrality and surjectivity on  $W^{-m}A$ : the formula is  $\mathbf{Z}_p$ -valued on the dense polynomial submodule, and  $\mathbf{Z}_p$  is closed in  $K$ , so its continuous extension is still  $\mathbf{Z}_p$ -valued.

Let  $x \in T^{-m}A$ . The function  $x$  has a pole among the  $p$ -th roots of unity only at  $q = 1$ . On each branch of  $u = q^p$ ,

$$\frac{dq}{q} = \frac{1}{p} \frac{du}{u}.$$

Summing the displayed branchwise change-of-variables identity shows that the  $p$  residues of  $\varphi(x) dq/q$  equals the residue of  $x dq/q$  at  $q = 1$ . Thus  $\text{Res}_p((\varphi - 1)x) = 0$ .

Finally,  $q \mapsto q^s$  fixes every  $p$ -th root of unity and is a local isomorphism there. Since  $d \log(q^s) = s d \log(q)$ , change of variables gives

$$\text{res}_{q=\zeta} \left( y(q^s) \frac{dq}{q} \right) = s^{-1} \text{res}_{q=\zeta} \left( y(q) \frac{dq}{q} \right),$$

which proves the last formula. □

For  $i \geq 2$ , the residue is compatible with the horizontal map. Namely, for  $[a] \in Q_i$ ,

$$(4.8) \quad \text{Res}_{i-1}^H(\bar{L}_i[a]) = (s^{i-1} - 1) \text{Res}_i^H([a]).$$

This follows by applying Lemma 4.8 to

$$P_i(W^{-i}a) = W^{1-i}L_i(a).$$

We also need one explicit class. Put

$$(4.9) \quad z_1 := (1 - p)^{-1} \left( d - p - \frac{p-1}{2} W \right) \in A.$$

**Lemma 4.9** (A residue-one class). *The class of  $z_1$  in  $Q_1$  lies in  $\ker(\bar{L}_1)$ , and*

$$\text{Res}_1^H([z_1]) = 1.$$

*Proof.* The required constant terms can be read off before any quotient is taken. Indeed,

$$d - p = \binom{p}{2}T + O(T^2), \quad \tilde{r}_1(d - p) - (d - p) = p \binom{p}{2}T + O(T^2),$$

while  $W = pT + O(T^2)$ . Hence the constant term of  $L_1(d - p) = (\tilde{r}_1(d - p) - (d - p))/W$  is  $\binom{p}{2} = p(p - 1)/2$ . On the other hand,

$$L_1(W) = p.$$

Since  $Q_0 = A/(\varphi - 1)A$  is identified with  $\mathbf{Z}_p$  by the constant term, equation (4.9) gives  $\bar{L}_1[z_1] = 0$ .

For  $m = 1$ , formula (4.7) says that the residue is the average of the values at the  $p$ -th roots. Since  $d = \sum_{r=0}^{p-1} q^r$ ,

$$\text{Res}_p \left( \frac{d - p}{W} \right) = 1 - p.$$

The term  $(p - 1)W/2$ , after division by  $W$ , is regular and has zero residue. Hence  $\text{Res}_1^H([z_1]) = 1$ .  $\square$

**Proposition 4.10** (The finite cokernel). *For  $i \geq 2$ ,*

$$\text{coker}(\bar{L}_i) \cong \mathbf{Z}_p/(s^{i-1} - 1).$$

*Proof.* Let

$$x_i = [W^{i-2}z_1] \in Q_{i-1}.$$

In the fractional model this is the same element  $W^{-1}z_1$ , so Lemma 4.9 gives

$$\text{Res}_{i-1}^H(x_i) = 1.$$

Equation (4.8) therefore induces a homomorphism

$$\overline{\text{Res}}_i^H : \text{coker}(\bar{L}_i) \longrightarrow \mathbf{Z}_p/(s^{i-1} - 1)$$

that sends the class of  $x_i$  to 1.

Apply (4.3) with  $k = i - 1$  and  $a = z_1$ . Since  $\bar{L}_1[z_1] = 0$ , there is  $b \in A$  with

$$L_1(z_1) = F_0(b).$$

The identity

$$W^{i-1}F_0(b) = F_{i-1}(T^{i-1}b)$$

shows that the second term on the right side of (4.3) vanishes in  $Q_{i-1}$ . Hence

$$(s^{i-1} - 1)x_i = 0 \quad \text{in } \text{coker}(\bar{L}_i).$$

By Lemma 4.7, the finite  $\mathbf{Z}_p$ -module  $\text{coker}(\bar{L}_i)$  has at most one generator. The residue of  $x_i$  is a unit, so its reduction is nonzero; Nakayama's lemma therefore shows both that the mod- $p$  cokernel is one-dimensional and that  $x_i$  is a generator. We obtain a surjection

$$\mathbf{Z}_p/(s^{i-1} - 1) \longrightarrow \text{coker}(\bar{L}_i), \quad 1 \longmapsto x_i.$$

Its composite with  $\overline{\text{Res}}_i^H$  is the identity. Both maps are therefore isomorphisms.  $\square$

**Theorem 4.11** (Direct cohomology of the  $q$ -Ore complex). *For every  $i \geq 1$ ,*

$$H^0(C_i^\bullet) = 0, \quad H^1(C_i^\bullet) \cong A/(d) \oplus \mathbf{Z}_p/(s^i - 1),$$

$$H^2(C_i^\bullet) \cong \begin{cases} 0, & i = 1, \\ \mathbf{Z}_p/(s^{i-1} - 1), & i \geq 2. \end{cases}$$

*The isomorphism in degree one is noncanonical.*

*Proof.* Combine Proposition 4.5, Corollary 4.6, and Proposition 4.10.  $\square$

**Corollary 4.12** (Cyclotomic syntomic cohomology). *Let  $R = \mathbf{Z}_p[\zeta_p] = A/(d)$ . For every  $i \geq 1$ ,*

$$\begin{aligned} H^0(\mathbf{Z}_p(i)(R)) &= 0, \\ H^1(\mathbf{Z}_p(i)(R)) &\cong R \oplus \mathbf{Z}_p/((p+1)^i - 1), \\ H^2(\mathbf{Z}_p(i)(R)) &\cong \begin{cases} 0, & i = 1, \\ \mathbf{Z}_p/((p+1)^{i-1} - 1), & i \geq 2. \end{cases} \end{aligned}$$

*The isomorphism in degree one is noncanonical as an isomorphism of  $\mathbf{Z}_p$ -modules.*

*Proof.* Corollary 3.4 gives a natural equivalence  $\mathbf{Z}_p(i)(R) \simeq C_i^\bullet$ . Apply Theorem 4.11 and use  $A/(d) = R$ .  $\square$

*Proof of Theorem 1.1.* The equivalence with the  $q$ -operator square is Corollary 3.4. Its direct cohomology calculation is Theorem 4.11, and  $A/(d) = R$ . These statements give all parts of Theorem 1.1.  $\square$

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